The Incidence and Real Activity Effects of Commercial Property Taxes: Firm-Level Evidence

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Abstract

Property taxation is widely viewed as a promising revenue source for developing countries, yet evidence on its effects on real economic activity and distributional effects remains limited. This paper exploits a natural experiment in Kampala, Uganda, where a property revaluation and mass enumeration increased tax liabilities in the city while surrounding areas remained unaffected. Using administrative data from corporate and employee tax returns in a difference-in-differences framework, I document three main findings. First, the reform increased business rental expenses by 9 percent, with an estimated citywide pass-through rate of 0.42, contrary to the neoclassical prediction that landlords should bear nearly the entire burden. Second, the reform reduced business profits, investment, employment, and survival probability, while increasing wage informality. Third, examining distributional consequences, I find that workers at the top of the earnings distribution were disproportionately affected, implying that the reform reduced within-firm wage inequality and was therefore progressive, a pattern consistent with skill downgrading within firms. Heterogeneity analysis reveals that financial constraints and distress play a role in explaining the results.

JEL codes: H22, H24, H32, H71.
Key Words: Property Taxes, Firm Survival, Employment, Wages.

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I. INTRODUCTION

As fiscal space shrinks in many developing countries, governments are exploring ways to strengthen existing revenue instruments and develop new sources to generate additional revenue. Property taxation is widely viewed as holding significant untapped potential, with the opportunity to raise substantial revenue by increasing compliance from currently low levels.[1] In this context, recent literature examines the effectiveness of alternative policies and administrative strategies in improving property tax compliance and revenues (e.g., Manwaring and Regan 2025; Flores et al. 2025; Bergeron et al. 2024). However, evidence on the efficiency cost of property taxation and its distributional consequences remains limited.

While the revenue potential of property taxes can strengthen local governments' capacity to provide public goods and services that support a conducive business environment, such taxes also raise costs for local firms. For owner-occupiers, the burden is direct. Firms operating in rented premises may instead face higher rents if landlords pass the tax on to tenants. Consistent with this channel, recent studies document substantial pass-through of property taxes to rental costs (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019). These higher costs may affect firms in many ways. In this paper, I study the incidence of commercial property taxes at the firm level and examine their effects on firm survival, revenues, employment, wages, and the distributional consequences.

I exploit a natural experiment in Kampala, Uganda, where variations in property tax liabilities were introduced across different locations within the city through a revaluation and mass property enumeration exercise that began in 2016/17. Prior to this exercise, the last valuation had been conducted in 2006, meaning that taxable property values had remained unchanged for a decade. During this period, numerous new properties had been established that were not captured in the old tax roll. The 2016/17 exercise led to the creation of a new property tax roll, which included additional properties and assigned higher rateable values to existing ones. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, although the marginal tax rate remained unchanged in both cases. Following the exercise, the total number of properties on the tax roll increased by 176 percent, while potential revenue more than tripled.

This exercise was implemented in a staggered manner, beginning with one division

[1] In Uganda, it is estimated that only 10–11 percent of properties pay their annual property tax liability, a rate similar to that observed in many other cities in low-income countries (Manwaring and Regan 2025).

and gradually extending to others over time. For context, the capital city is divided into five administrative divisions. I exploit this geographical variation using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with firms located outside the capital but within the Greater Kampala Metropolitan Area serving as never-treated units. This strategy allows me to credibly estimate the causal effects of increases in property tax liabilities on firm-level outcomes. The identifying assumption is that, in the absence of the reform, outcomes for firms in treated and untreated locations would have evolved similarly, i.e., the parallel trends assumption. I test this assumption using an event-study analysis. The analysis utilizes administrative tax data from corporate income and employee tax returns. These novel firm-level panel datasets span nine years (2013/14 - 2021/22) and were recently compiled by the Uganda Revenue Authority with support from UNU-WIDER.

As a "first stage" of the empirical analysis, I estimate the effect of the reform on business rental expenses and calculate the pass-through rate. I focus on firms in the balanced panel that reported paying rent in the year prior to the reform, estimating the effect of the reform on rent expenses scaled by their pre-reform values. I find that the reform led to a 9 percent increase in business rental expenses. This reduced form estimate implies that a UGX 1 increase in potential citywide property tax liabilities is associated with a UGX 0.42 increase in total rents paid by firms.[2] Interpreted as an incidence measure at the city level, this implies that approximately 42 percent of the aggregate commercial property tax burden is transmitted to firms through higher rents. Given that only about 65 percent of potential commercial property tax liabilities are ultimately collected, this estimate further implies that a UGX 1 increase in collected citywide property tax revenue is associated with a UGX 0.65 increase in total rents paid by firms. This result contrast the neoclassical view which asserts that the burden property taxes should fall almost entirely on the landlords. To explain this contrast, I propose a framework that embeds a key institutional feature of selective enforcement: the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity, and lower when the unit is unoccupied.

I then analyze how the reform affected business revenues and profits, focusing on a panel of active firms. Evidence points to a short-term positive effect on revenues that quickly dissipates, likely reflecting firms' attempts to pass the burden to consumers through higher product prices, although I cannot directly observe prices. Overall, no significant

[2] Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, it is not possible to construct a firm-level measure of tax pass-through.

average effects are found on the revenues. However, the reform reduced firms' profits by 3.3 percentage points relative to pre-reform revenues, while investment reduced by 2.7 percent.

To further understand firm dynamics, I examine the effects of the reform on firm survival. The survival outcome is constructed following the approach of Saez et al. (2019), restricting the sample to firms present in 2015/16 (the year prior to the reform) that reported non-zero turnover, regardless of whether they operated in other years. Firms are assigned a turnover value of zero in years in which they do not operate. The outcome variable is binary, equal to one if the firm operated in that year and zero otherwise. Consistent with Saez et al. (2019), the sample is reweighted using the nonparametric reweighting method developed by DiNardo et al. (1996) (DFL reweighting). The results indicate that the reform reduced firms' survival probability by 3.2 percent. Scaling this reduced-form effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 27 additional firm exits citywide. Further scaling by the share of commercial property tax liabilities that is ultimately collected implies that a $1 million increase in collected citywide property tax revenue is associated with approximately 42 additional firm exits citywide.

A potential threat to my results is that they could be compounded by evasion of corporate income tax (CIT). To address this, I use a cross-tax validation strategy which classifies a case as exit if and only if the firm disappears from CIT and has no footprints in the employee tax returns. I further impose a stricter definition of exit, requiring no CIT return in year t and no record in employee tax returns in both years t and $t + 1$. Both of these restrictions show that the results are not driven by evasion responses. Moreover, heterogeneity analysis reveals that the negative survival effects are strongest among firms with no corporate income tax liability (i.e., loss-making firms). As an additional robustness check, the results are similar when the sample is restricted to firms that were present and reported positive turnover in the first year of the sample, rather than to firms in the year immediately prior to the reform.

Using the same sample, I analyze employment effects, coding employment as zero following firm exit and normalizing it relative to its pre-reform level to facilitate interpretation in percent changes. On average, the reform reduced employment by 4.5 percent among firms in treated areas relative to untreated areas. Scaling this reduced-form estimate by the aggregate increase in potential property tax liabilities implies that a $1 million rise in aggregate liabilities is associated with approximately 206 fewer jobs citywide.[3]

[3] As with pass-through to rents, data limitations preclude calculation of firm-level employment elasticities with respect to property tax liabilities.

Adjusting for the collection rate increases this estimate to 317 fewer jobs per $1 million in collected commercial property tax revenue. In a sample of firms active throughout the entire period, employment declines by 4 percent, indicating that the overall reduction in employment is not driven solely by firm exit but also by within-firm adjustments among continuing firms. As a robustness check, I re-estimate the employment effects using log and inverse hyperbolic sine transformation as well as levels.[4] Both the event-study plots and average effects are consistent with those obtained using normalized outcomes.

Consistent with the employment results, I find that the firms' wage bill declines by 5.5 percent following the reform. The results further reveal the reform led to an increase in informal wages by 3.3 percentage points from a pre-reform mean of 10.4 percent, suggesting that part of the observed decline in formal employment reflects jobs transitioning to informal arrangements rather than outright job destruction. This partly accounts for the large estimated employment effects.

Heterogeneity analysis reveals that both survival and employment effects are stronger and statistically significant only among tenant firms, and not among owner-occupier firms. This pattern is consistent with the substantial pass-through to business rental expenses, as well as with the fact that owner-occupier firms tend to be larger and better able to absorb higher property tax liabilities by adjusting other margins. Additional heterogeneity analysis shows that the effects are stronger among less profitable and more highly leveraged firms, pointing to financial constraints and financial distress as key underlying mechanisms.

Finally, I examine the effects of the reform on wages and their distributional consequences by assessing which types of workers are most affected. The sample is restricted to active firms with at least five workers in the pre-reform period to ensure adequate support across the wage distribution. First, I analyze how within-firm wage quantiles evolve before and after the reform in treated firms relative to untreated firms. Second, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the 75th percentile of wages declined by 0.18 percentage points, relative to the pre-reform wage bill, in treated firms compared to untreated firms, while there is no statistically significant effect on the 25th and 50th percentiles, despite negative point estimates. I document a similar pattern when using four instead of three quantiles. These results suggest that the reform reduced within-firm wage inequality and was therefore progressive. Consistent with the above findings, I find that the share of workers in the

[4] The log transformation is log(1+employment) since employment can be zero in some years.

top quartile of the pre-reform wage distribution declined by 2.8 percentage points relative to pre-reform total employment, while the share of workers in the 50th–75th percentile increased by 2.2 percentage points. These findings are consistent with skill downgrading within firms.

Event-study analyses show that the results on business rental expenses, firm survival, employment, and wages are not driven by pre-trends. The event-study plots indicate that pre-period coefficients are statistically indistinguishable from zero, suggesting the absence of differential pre-trends and that outcomes would have evolved similarly in treated and untreated areas in the absence of the reform.

This paper contributes to two distinct strands of the literature. First, it relates to research on how local taxes affect business decisions. Recent studies show that state and local tax policies in the United States influence business location choices (Fajgelbaum et al. 2019; Giroud and Rauh 2019; Rohlin et al. 2014) and employment decisions (Thompson and Rohlin 2012; Giroud and Rauh 2019). Far less attention, however, has been devoted to understanding how local taxes, particularly property taxes affect real activity and their distribution consequences. Moreover, the existing literature largely examines local sales taxes in developed-country contexts. By contrast, this paper focuses on property taxes in a developing-country setting. The effects of property taxes are likely to differ from those of sales taxes because the incidence falls on a different base. For example, the extent to which property taxes affect firms operating in rented premises depends on the pass-through rate from landlords to tenants. Most closely related to this paper is the recent work by Enami et al. (2023), which examines how school property taxes affect both the stock and flow of establishments in a location. However, their analysis focuses on a single type of property, schools, finding null effects.

I contribute to the literature by showing that commercial property taxes broadly affects business survival, employment, and wages. While no significant effects are found among owner-occupier businesses, consistent with Enami et al. (2023) on school property taxes, substantial effects are uncovered among tenant businesses, driven by high tax pass-through. This implies that analyses focusing solely on property owners may substantially understate the true impact of property taxation on real activity. This paper also highlights financial constraints and financial distress as key mechanisms underlying these effects.

Second, this paper contributes to the long-standing literature on property tax incidence (see England (2016) and Oates and Fischel (2016) for reviews). While earlier work found low pass-through of property taxes to rent (Carroll and Yinger 1994), more recent studies have documented substantial pass-through to rental costs (Watson and Ziv 2024; Baker 2024), particularly when the supply of housing is elastic (Löffler and Siegloch

2021). Prior research, however, has focused primarily on rental housing. This paper provides evidence on pass-through in the context of commercial properties.

The remainder of the paper is organized as follows. Section II describes the institutional context, including property taxation arrangements in Uganda and details of the revaluation and property enumeration reform. Section III describes the corporate income tax and employee tax return data used in the analysis and presents descriptive statistics. Section IV outlines the identification strategy. Section V presents and discusses empirical evidence on the effects of the reform on business rental expenses and the associated theoretical contrasts. Section VI presents the empirical results on the effects of the reform on various dimensions of business activity, including revenues, firm survival, and employment, as well as the heterogeneity analysis. Section VII examines the effects on wages and the associated distributional implications. Section VIII concludes.

II. CONTEXT

II.A. Property Taxation in Uganda

Property taxes in Uganda are governed by the Local Government (Rating) Act, 2005, which allows local jurisdictions to levy rates, based on the rateable value of any property within their jurisdiction. The tax is levied on all immovable property, including buildings or structures of any kind that are commercially managed (e.g., rented residential houses, rented shops, factories, hotels). It also applies to owner-occupied properties used for business purposes but excludes vacant sites and owner-occupied residential properties.[5] Local governments have the discretion to levy rates ranging from a minimum of 2,000 Uganda Shillings (approximately $0.60) up to 12 percent of the rateable value, where rateable value is defined as the assessed property value less a conservancy allowance of 22 percent. In Kampala Capital City Authority (KCCA), the property tax rate is set at 6 percent of the rateable value.

Currently, most cities (including the capital) and municipalities in Uganda use a market-value approach in which property rates are applied to the estimated annual rental value or business income value (Bakibinga and Ngabirano 2019). The law requires each local government to produce an initial valuation list and update it at least once every five years.

[5]The full list of exempted properties include: official residence of the president; official residences of cultural/traditional leaders; public worship places; cemeteries and crematoriums; charitable and educational institutions; outdoor sports and recreational facilities; local council-owned properties; property owned by privileged organizations under Diplomatic Privileges Act; International Treaty Tax-exempt organizations; and property belonging to any institution with which Government has contractual obligation not to levy fees and tax.

In practice, however, these lists are often not updated on time, largely due to a shortage of qualified valuation surveyors in both the private and public sectors (Bakibinga and Ngabirano 2019) and inadequate resources.

II.B. Property Revaluation, Enumeration, and Tax Collection in Kampala

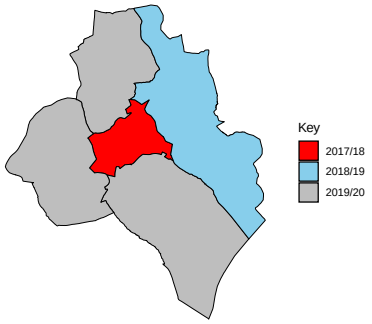
In an effort to increase revenue, the Kampala Capital City Authority (KCCA) initiated a revaluation and enumeration of properties beginning in the financial year 2016/17. Prior to this, the last valuation had been conducted in 2006, implying that the taxable values of properties had remained unchanged for a decade. Over the same period, numerous new properties had been developed that were not included in the old tax roll. The number of properties on the tax roll increased by 176 percent, from 127,021 to 223,329, while the average property tax liability increased by 90 percent after the exercise. The revaluation and enumeration exercise was therefore expected to substantially increase revenue for KCCA, where property tax had accounted for 35 percent of own-source revenue prior to the exercise. During the mass enumeration and surveying, KCCA collected detailed information on ownership, location (including GIS coordinates), and property attributes for valuation purposes. Valuation was conducted primarily by professional valuers, although in some cases it relied on regression model predictions.

This exercise was conducted in a staggered manner, starting with one division and gradually extending to others over time. Figure 1 shows the rollout of the new tax roll across the five divisions of the capital city, following the revaluation and enumeration exercise. The different colors in the figure represent the various phases of the exercise, as indicated in the legend.

The exercise resulted in the introduction of a new property tax roll that included more properties and assigned higher rateable values to existing ones. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, i.e., from zero to a positive tax liability, although the marginal tax rate remained unchanged in both cases. Figure 2 presents the number of properties and the revenue from each of the five divisions under both the old and new tax rolls. All divisions experienced an increase in the number of properties, with the number more than doubling in some divisions (Figure 2, Panel A). Similarly, potential revenue per property— a proxy for tax liability, increased across all divisions, with some experiencing more than 180% increase (Figure 2, Panel B).

Property Tax Collection:— After the valuation, initial valuation lists are displayed in hard copy at the respective division offices, with additional publication through the press, radio announcements, and local leaders. Property owners are then given one month to

Panel A: Kampala Capital City



Panel B: Greater Kampala Metropolitan Area

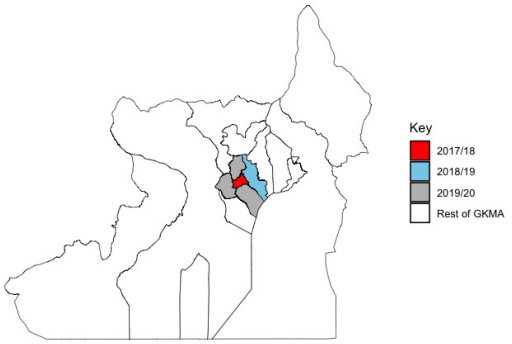
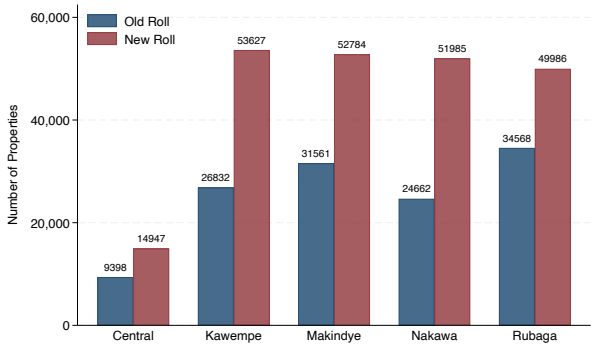


Figure 1: Kampala Capital City and Surrounding Areas

Notes: Panel A shows the five divisions of Kampala Capital City and the different phases of the rollout of the new property tax roll, as indicated in the legend. Panel B shows the Greater Kampala Metropolitan Area (GKMA), which includes counties in the neighborhood of the capital city. Firms in these locations serve as the never-treated units in the identification strategy, as these areas did not experience similar changes in property tax enforcement during the study period.

Panel A: Number of Properties



Panel B: Potential Revenue Per Property

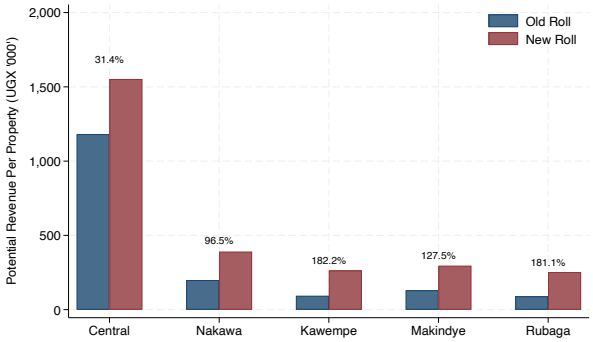


Figure 2: Number of Properties and Potential Property Tax Revenue

Notes: Panel A shows that number of properties in the old and new roll across the five divisions. Panel B shows the potential revenue per property before and after the reform. The data is extracted from Ahabwe et al. (2020) and Bakibinga and Ngabirano (2019).

file objections, after which the valuation court convenes to hear all appeals. Decisions on the appeals may result in either an increase or a decrease in tax liability. Final valuation lists are then generated, and bills and demand notes are printed and delivered to property owners. Although the Authority is not able to physically deliver demand notes to every property owner, the digital system, eCitie, sends SMS and email notifications to all own-

ers of the billed property.[6] KCCA is also legally authorized to issue a demand notice to a tenant if the property owner is unavailable, and the tenant is legally entitled to deduct the tax payment from their rent.

Billing for all properties included on the final valuation lists occurs at the beginning of the financial year (July 1). Taxpayers have until December 31 to pay their property taxes on time. They may request payment plans, which are officially limited to a maximum of four equal installments. If payment is not made in time, KCCA may issue a demand notice requiring the taxpayer to settle the outstanding rate, along with any accrued interest, within two months of the notice being served. Interest begins accruing 30 days after the end of the financial year, at a rate of 2 percent per month, for as long as the liability remains unpaid. Payments can be remitted through the bank or via mobile money.

III. DATA AND DESCRIPTIVE STATISTICS

III.A. Data Sources and Sample Construction

The paper utilizes administrative tax data from corporate income and employee tax returns as its primary datasets. The corporate income tax return (CIT) data comprise a firm-level panel covering the universe of firms registered for corporate income tax in Uganda, spanning the period 2013/14 – 2021/22. The employee tax return data cover the universe of individuals in formal employment over the same period. These are newly compiled dataset launched in 2022 by the Uganda Revenue Authority and UNU-WIDER.

The corporate income tax data contain annual balance sheets and profit and loss accounts, which allow me to observe detailed measures of business activity, including firms' revenues, profits, and assets. All monetary values are recorded in Uganda shillings. To ensure comparability over time, I deflate them to constant 2016/17 values using the consumer price index (CPI). The data also include information on firms' sectors of operation and location. Location information is particularly important for defining treatment, as discussed later in the paper. Because the corporate income tax data do not contain information on employment, I link them to the employee tax return data to construct a measure of formal employment by counting the number of workers filing under each firm. Since employee tax returns are filed monthly while corporate income tax returns are filed annually, I compute average employment over the twelve months of each year.

Sample Construction:— The full dataset from corporate income tax returns comprises 404,527 firm–year observations. However, some firms report zero turnover throughout

[6] Most of the information on the valuation and billing process was obtained from (Manwaring and Regan 2019).

the entire sample period. After dropping these firms, the dataset is reduced to 299,683 firm–year observations, of which 81,162 (27 percent) form a balanced panel comprising 9,018 unique firms. The sample is further restricted to firms located in the Greater Kampala Metropolitan Area (GKMA) to ensure that never-treated firms are geographically as close as possible to treated firms. Because the reform affects only firms in the capital city, firms in surrounding locations, the rest of the GKMA (Figure 1, Panel B), serve as a plausible comparison group. This restriction yields 213,185 firm–year observations, representing 71 percent of firms that ever reported positive revenues during the sample period. Of these, 60,984 observations form a balanced panel comprising 6,776 unique firms, equivalent to 75 percent of the balanced-panel observations in the full dataset.

Additional sample restrictions vary by section or analysis. For the analysis of the effect on business rental expenses, the sample is restricted to firms that reported paying rent in the pre-reform period, as this outcome is normalized by its pre-reform value. For the baseline analyses of firm survival and employment effects, the sample is restricted to firms that operated and reported positive revenues in the pre-reform year.

III.B. Descriptive Statistics

Figure 3 plots the industry composition and size distribution across treated and comparison locations for firms that were active in the pre-reform year. The figure reveals substantial overlap along these dimensions. The size distribution is left-skewed: the majority of firms have annual revenues below UGX 150 million and fewer than five formal employees. However, the distribution is not systematically different for firms inside versus outside the city. Appendix Figure A.1 shows a similar overlap using density plots for revenues and assets. There are some notable differences in industry composition across sectors. For example, in retail and wholesale trade, the share of firms inside the city exceeds that outside the city by a considerable margin, while the reverse holds for the education sector. I flexibly account for differences in size and industry composition using fixed effects in the analysis. There is sufficient overlap in both dimensions to allow for comparisons of firms within the same industry and size bins across treated and comparison locations.

Table 1 reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. The average firm in the universe of corporate income tax (CIT) filers has annual revenues of UGX 556 million and total assets of UGX 267 million, of which 70 percent are fixed assets. Average revenues for treated and comparison firms exceed those in the CIT universe, but are broadly comparable across the two groups. In contrast, firms in comparison group hold more assets on average than treated firms, sug-

Figure 3: Firm Size Distributions and Industry Composition

Notes: This figure plots the industry and size mix of pre-reform active firms located inside (treated) and outside (comparison) the city. Each graph's bars sum to 1 within each location. Panels A and B show the distribution of revenues and employment, respectively. Panel C shows the industry composition of firms in the sample following the International Standard Industrial Classification of All Economic Activities (ISIC, Rev.4). Utilities combines electricity, gas, steam, and air conditioning supply with water supply, sewerage, waste management, and remediation activities. Administration combines administrative and support service activities with public administration & defence and compulsory social security. I also exclude ISIC categories for activities of households as employers; undifferentiated goods- and services-producing activities of households for own use; and activities of extraterritorial organizations and bodies, as these account for negligible shares in the sample.

gesting that firms in the city (treated locations) are more efficient in utilizing assets to generate revenue. Similarly, average formal employment is higher among comparison firms than among treated firms. To account for these size-related differences, I include in the regressions time-invariant size bins defined by pre-reform employment. Nonetheless, robustness checks confirm that results are insensitive to the inclusion or exclusion of size fixed effects.

Regarding renting or owning land and buildings, the average firm in the CIT universe pays annual rent of UGX 6.2 million and owns land and buildings worth UGX 72 million. Comparison firms have lower average rent expenses and are less likely to be renters (report positive rent) compared to treated firms. In contrast, comparison firms have higher average values of land and buildings and are more likely to own property (report positive values for land and buildings). To understand the implications of these differences, I disaggregated all my main results by whether firms are tenants or owner-occupiers.

Lastly, I present statistics on variables related to the financial situation of firms. Firms with better financial situations may respond differently to cost increases associated with property tax liabilities. Table 1 shows that the share of loss-making firms is balanced at 23 percent across comparison and treated firms and slightly lower in the CIT universe. Similarly, average profit margins (EBIT/revenue) are similar across comparison and treated firms. Treated firms have higher liquidity (cash-to-assets) relative to comparison firms. While leverage is slightly higher in the CIT universe at 41 percent, it is balanced across treated and comparison firms. Broadly, comparison and treated firms exhibit similar pre-reform financial situations along the dimensions of profitability and leverage.

IV. IDENTIFICATION STRATEGY

I use difference-in-differences design to estimate the causal impact of property tax enforcement on business rental expenses, business activity, and wages. The identifying variation comes from comparing outcomes of firms located in Kampala versus those located in surrounding areas that did not experience a change in property tax enforcement and liability during the sample period. The identifying assumption is that, absent the reform, average outcomes for firms in treated and untreated areas would have evolved similarly (i.e., parallel trends assumption). I test this assumption using event-study analysis.

Traditionally, one would estimate the average and dynamic effects of the reform using Two-Way Fixed Effects (TWFE) difference-in-differences regressions, as in equations 1 and 2, respectively.

$$y_{it} = \alpha_g + \alpha_t + \beta D_{it} + \gamma X_{it} + \epsilon_{it}, \tag{1}$$

Table 1: Summary Statistics

	(1) Universe	(2) Ever-Treated	(3) Comparison Group
Revenues	556.09	614.96	629.81
	(1,232.30)	(1,294.14)	(1,328.00)
Total Assets	267.43	284.87	361.35
	(595.70)	(620.41)	(696.65)
Fixed Assets	188.27	190.70	294.15
	(461.14)	(468.84)	(568.66)
Formal Employees	5.16	5.15	7.45
	(10.69)	(10.43)	(13.00)
Rent Expenses	6.15	8.39	4.43
	(12.84)	(14.83)	(11.27)
Landing and Buildings	72.33	63.74	130.18
	(209.02)	(202.33)	(270.43)
$\mathbb{I}\{\text{Rent Expense} > 0\}$	0.44	0.50	0.34
	(0.50)	(0.50)	(0.47)
$\mathbb{I}\{\text{Landing and Buildings} > 0\}$	0.20	0.15	0.31
	(0.40)	(0.36)	(0.46)
$\mathbb{I}\{\text{Loss Making}\}$	0.20	0.23	0.23
	(0.40)	(0.42)	(0.42)
Profit Margins	0.15	0.13	0.14
	(0.31)	(0.31)	(0.33)
Cash-to-Assets	0.17	0.17	0.13
	(0.22)	(0.23)	(0.20)
Debt-to-Assets	0.41	0.35	0.36
	(0.36)	(0.35)	(0.35)

Note: Reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. Standard deviations are reported in parentheses. Column 1 presents statistics for the CIT universe, while the remaining columns present statistics for different groups according to their treatment status. The ever-treated group consists of firms located in Kampala City, whereas the comparison group consists of firms from the rest of the GKMA. All monetary values are denominated in Uganda Shillings (UGX), constant 2016/17.

$$y_{it} = \alpha_g + \alpha_t + \sum_{\substack{e=-5 \\ e\neq -1}}^{4} \beta_e D_{it}^e + \gamma X_{it} + \nu_{it}, \tag{2}$$

where y_{it} is the outcome variable, i.e., business rental expenses, business activity mea-

sures, and wages. Detailed description of outcome measurement is presented in the respective sections, following this section. D_{it}^{e} is a time-varying indicator variable that is equal to one starting in the year firm i's location received the reform and zero otherwise. α_g and α_t are group fixed effects and year fixed effects. X_{it} is a vector of controls.

However, recent literature shows that under staggered adoption, TWFE regressions deliver consistent estimates only under homogeneous treatment effects. In the presence of heterogeneous effects across cohorts or over time, TWFE coefficients are generally biased and difficult to interpret (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Callaway and Sant'Anna 2021; Borusyak et al. 2024). Bias in TWFE estimators arises from negative weighting induced by forbidden comparisons, whereby already-treated units are used as controls for later-treated units (Borusyak and Jaravel 2018). Heterogeneous effects are plausible in this setting for two reasons. First, firms' responses to changes in property tax enforcement and liability may evolve over time as they reoptimize along different margins. Second, changes in property tax liability vary across cohorts (Figure 2), implying treatment effects may differ by cohorts.

To circumvent the challenges associated with TWFE, I estimate treatment effects using the doubly robust estimator of Callaway and Sant'Anna (2021), using the not-yet-treated units as the comparison group. All the results are robust to using the never-treated firms as the comparison group. I also show that the main difference-in-differences and event-study estimates are robust when estimated using the estimators introduced in Sun and Abraham (2021) and Borusyak et al. (2024). Callaway and Sant'Anna (2021) show that this estimator is consistent in the presence of cohort-specific treatment effects. In addition, their approach allows for estimation and inference under conditional parallel trends, i.e., parallel trends assumption holds only after conditioning on observed covariates.

Rather than estimating dynamic effects via the event-study regression in equation 2, Callaway and Sant'Anna (2021) propose an aggregation scheme to highlights treatment effect heterogeneity by exposure length while avoiding the drawbacks associated with TWFE, as in equation 3.

$$\beta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{I}\{g + e \leq \mathcal{T}\}\, \mathbb{P}(G = g \mid G + e \leq \mathcal{T})\, ATT(g, g + e), \tag{3}$$

where, e denotes event time (the number of periods since initial treatment), G denotes the first period a unit is treated, and $ATT(g,t)$ is the average treatment effect for units who are members of a particular group g at a particular time period t, i.e., $ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid X, G_g = 1]$. $\beta_{es}(e)$ is therefore the average effect of participating in the treatment e periods after the treatment was adopted, averaging across all cohorts

observed at event time e.

In estimation, I recognize that the the roll out of the reform was not randomly assigned across groups. The city authority first targeted divisions with high-value properties, namely Central Division, followed by Nakawa (Figure 2, Panel B). Moreover, the never-treated firms are located outside the capital city and may differ systematically from ever-treated firms. As shown in Table 1, firms in the capital differ on observables from those outside the city. Accordingly, I include baseline controls, size bins, sector, and accounting month, in the regression specifications. I also present event-study evidence indicating no differential pre-trends between treated and comparison groups.

V. EFFECTS ON RENT EXPENSES

V.A. Empirical Evidence

Given that a substantial share of businesses in the sample (about 45%) report paying rent in the pre-reform year, it is important to examine whether property taxes affect business rental costs, as a "first stage" of the empirical analysis. According to the neoclassical view, the burden of property taxation should fall on the least elastic factor, implying that landlords ultimately bear the tax incidence. If this prediction holds, property taxes should have no direct effect on business rental expenses. There is, however, limited empirical evidence testing this prediction in the context of commercial properties. Recent work focusing on rental housing provides evidence that contrasts with this prediction (e.g., Baker 2024).

I test whether the neoclassical prediction holds in the context of commercial properties. I consider firms in the balanced panel that reported paying rent in the year prior to the reform and estimate the effect of the reform on rent expenses, scaled by their pre-reform values. The results are presented in Figure 4. Prior to the reform, rent expenses for firms in locations that experienced an increase in property tax liability and those in unaffected areas followed similar trends, supporting the identifying assumption of parallel trends. Following the reform, firms in treated locations experienced a differential increase in rent expenses relative to those in untreated areas. The average effects presented Table 2 shows that the reform led to a 9 percent increase in business rental expenses (column 4).

Implied Pass-Through Rate:— To interpret the economic magnitude of the reform's effects on rental expenses, I compute an implied pass-through rate using the reduced-form estimate. Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. Moreover, multiple firms may operate within the same property. As a result, it is not possible to

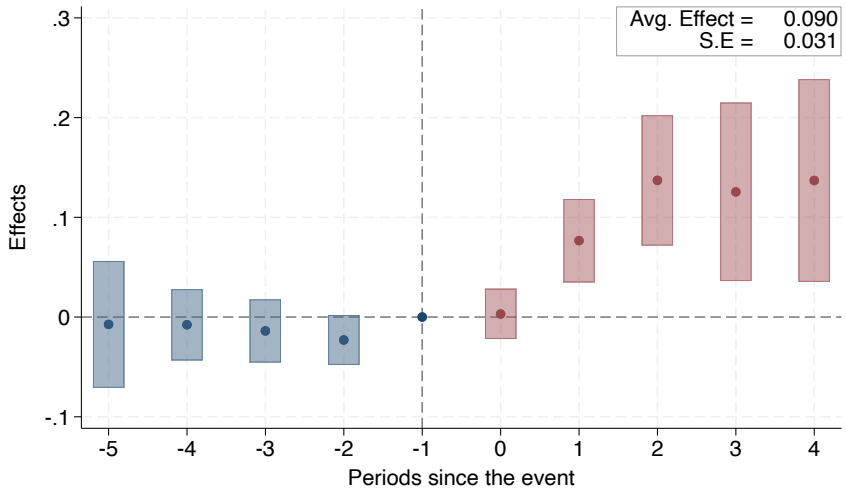


Figure 4: Effects of Property Taxes on Business Rental Expenses

Notes: The figure shows the dynamic effects of property taxes on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Appendix Figure A.2 shows the event-study plot using never treated units as the comparison group. Standard errors are clustered at the firm level.

construct a firm-level measure of tax exposure or pass-through. Given these limitations, I estimate pass-through at the citywide level. Conceptually, this approach treats the reform as an aggregate shock to the local cost of supplying commercial property and asks what fraction of the resulting increase in potential property tax revenue is transmitted to firms through higher rents. I operationalize this by scaling the reduced-form increase in rents paid by firms by the aggregate increase in citywide property tax liabilities induced by the reform. Formally, I define the citywide pass-through rate as: $\frac{\hat{\beta} * \bar{R} * N}{\Delta T^{\text{City}}}$, where $\bar{R}$ denotes the average pre-reform rent expense, N is the total number of firms operating in the city prior to the reform, and ΔT^{City} is the reform-induced change in citywide property tax liabilities. This measure captures the share of the aggregate property tax increase that is ultimately borne by firms through higher rents. Importantly, it should not be interpreted as a structural, firm-level pass-through rate. Instead, it represents an incidence measure that aggregates across heterogeneous exposure to the reform, endogenous firm responses, and any general equilibrium effects operating through the local rental market.

Results in Table 2 (second row from the bottom) indicate that a UGX 1 increase in potential citywide property tax liabilities is associated with a UGX 0.42 increase in total

rents paid by firms. Interpreted as an incidence measure at the city level, this suggests that approximately 42 percent of the aggregate commercial property tax burden is transmitted to firms through higher rents. Given that only about 65 percent of potential commercial property tax liabilities are ultimately collected, this estimate implies that a UGX 1 increase in collected citywide property tax revenue is associated with a UGX 0.65 increase in total rents paid by firms.

While not directly comparable to property-level pass-through estimates in the housing literature, the citywide incidence measure estimated here is surprisingly similar in magnitude to recent studies of property tax pass-through in rental housing markets. These studies typically estimate pass-through at the property or unit level and, in some cases, embed the analysis in a general equilibrium framework that allows for endogenous adjustment in rents, occupancy, and housing supply, finding pass-through rates ranging from \$0.50 to \$1.39 per \$1 of tax (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019). By contrast, my estimate captures a reduced form incidence measure at the city level, aggregating across heterogeneous properties, landlords, and tenant firms, and abstracts from equilibrium reallocation across locations. Despite these differences in level of aggregation, empirical design, and the study context, the similarity in magnitudes is informative and suggests that a substantial share of the commercial property tax burden is transmitted to firms through higher rents.

My results are consistent with the "old view" of property tax incidence, which asserts that the portion of the tax falling on land is borne entirely by owners, since the supply of land is perfectly inelastic, while the portion falling on improvements—i.e., structures—can be shifted to final consumers and occupants, since the supply of capital to the construction industry is relatively elastic (Oates and Fischel 2016). In my setting, it is plausible to assume that the entire tax falls on improvements rather than on land itself, since idle land is exempt.

V.B. Theoretical Framework

The result that higher property tax liabilities lead to an increase in business rent expenses contrasts with the predictions of neoclassical theory, which suggests little to no pass-through of property taxes (Mieszkowski 1972; Zodrow 2001). Underlying the neoclassical prediction is the idea that demand for space in a location is elastic, since other jurisdictions offer comparable commercial units, while supply is inelastic. In that setting, the property tax falls on the immobile factor and landlords absorb the entire burden because any attempt to raise rents would drive tenants to competing locations.

A necessary condition for this result, however, is that the property tax is a cost of

Table 2: Effects of Property Tax on Business Rental Expenses

	(1)	(2)	(3)	(4)
Average Effect	0.069**	0.096***	0.097***	0.090***
	(0.030)	(0.031)	(0.031)	(0.031)
Sector of Operation	N	Y	Y	Y
Pre-reform Size	N	N	Y	Y
Other Controls	N	N	N	Y
Pre-reform Mean	35.46	35.46	35.46	35.46
Implied Pass-Through Rate	0.322	0.447	0.452	0.423
Observations	30798	30798	30798	30798

Note: The table presents average effects of the reform on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. Each column reports estimates from a different specification with an alternative set of controls. Under "Other Controls," I include dummies for accounting month, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row presents citywide pass-through calculated from the expression: $\frac{\hat{\beta} * \bar{R} * N}{\Delta T^{\text{City}}}$, where $\bar{R}$ denotes the average pre-reform rent expense, N is the total number of firms operating in the city prior to the reform, and ΔT^{City} is the reform-induced change in citywide property tax liabilities. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level. Pre-reform mean is measured in millions of Uganda Shillings.

owning the property, not a cost of *renting it out*. If the tax is enforced universally, the landlord owes the same liability whether the property is occupied or vacant, and the tax is effectively a lump sum that cannot influence the landlord's pricing decision. But if the tax is enforced selectively—collected only when the property is occupied and generating observable economic activity—then renting out the property triggers the tax liability. In this case, the tax functions as a marginal cost of supplying space to the rental market, and standard incidence logic predicts positive pass-through to tenants.

To rationalize my findings, I develop a simple framework that predicts non-zero pass-through and that does not depend on the elasticities of demand and supply in the standard sense. The framework embeds a key institutional feature of property tax enforcement in developing countries: *selective enforcement*, where the probability of enforcing, and therefore collecting, property taxes is lower on vacant properties than on occupied ones. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use and generating income (Franzsen and McCluskey 2017). When administrative resources are scarce, targeting occupied properties is a cost-effective strategy: occupied properties are easier to identify, their owners are easier to locate, and the revenue yield per enforcement action is higher because an active tenant signals ability to pay.

Setup:— Consider a landlord who owns one unit of commercial space and must choose the rent R to charge. If the unit is occupied at rent R, the landlord earns rental income R and pays the property tax liability $T = \tau V$ with certainty, yielding a payoff of $R - T$.[7] If the unit remains vacant, the landlord earns no rental income but may still face the tax: the tax is enforced on the vacant unit with probability $p_e \leq 1$, yielding an expected payoff of $-p_e T$. [8]

The probability that the unit is occupied depends on the rent charged. Let $q(R)$ denote the probability that the unit is filled at rent R, with $q'(R) < 0$ (higher rents reduce the likelihood of finding a tenant) and $q(0) \leq 1$. Under the assumption of a unit mass of potential tenant businesses, $q(R)$ also represents the aggregate demand for commercial space, i.e., the share of businesses willing to pay rent R rather than take their outside option.[9]

Landlord's Maximization Problem:— The landlord sets rent to maximize his expected utility:

$$\max_R \; q(R)\,(R - T) + (1 - q(R))\,(-p_e T) \tag{4}$$

The first term is the expected payoff if the unit is filled: the landlord earns $R - T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1 - q(R)$.

Expanding (4) and dropping the terms that do not depend on R, the landlord effectively maximizes:

$$\max_R \; q(R)\,[R - (1 - p_e)T] \tag{5}$$

This has a clear economic interpretation: the landlord behaves as if facing a per-unit cost of $(1 - p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero— the landlord pays T regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax T— each occupied unit triggers the entire tax bill.

[7] τ is the property tax rate and V is the assessed value.

[8] In a setting with imperfect compliance, even when the unit is occupied, as is common in many developing countries, p_e can be interpreted as the relative probability of compliance when the unit is vacant compared to when it is occupied. If we assume landlords are more likely to comply when the unit is occupied compared to when it's vacant, then $p_e < 1$.

[9] This equivalence holds when each potential tenant independently decides whether to rent the unit. If tenants have heterogeneous reservation values for the location, $q(R)$ is the fraction whose reservation value exceeds R.

The first-order condition of (5) is:

$$q'(R)\,[R-(1-p_e)T]+q(R)=0 \tag{6}$$

Solving for R:

$$\boxed{R^*=(1-p_e)T-\frac{q(R^*)}{q'(R^*)}} \tag{7}$$

The optimal rent has two components. The first, $(1-p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

Characterizing the Pass-through Rate:— Defining $c \equiv (1-p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent R^* as a function of the effective cost c, and hence of T. Define the function:

$$G(R,c)\equiv q'(R)(R-c)+q(R)=0 \tag{8}$$

By the implicit function theorem we have:

$$\frac{dR^*}{dc}=-\frac{-q'(R^*)}{q''(R^*)(R^*-c)+2q'(R^*)}=\frac{q'(R^*)}{q''(R^*)(R^*-c)+2q'(R^*)} \tag{9}$$

From the first-order condition (6), we can substitute $R^*-c=-q(R^*)/q'(R^*)$, we can write (9) as:

$$\frac{dR^*}{dc}=\frac{1}{2-\dfrac{q(R^*)\,q''(R^*)}{[q'(R^*)]^2}} \tag{10}$$

Since $c=(1-p_e)T$ and hence $dc/dT=(1-p_e)$, the pass-through rate with respect to the tax liability is:

$$\boxed{\rho=\frac{dR^*}{dT}=\frac{dR^*}{dc}\cdot\frac{dc}{dT}=\frac{1-p_e}{2-\theta}}, \tag{11}$$

where $\theta \equiv \frac{q(R^*)\,q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by Weyl and Fabinger (2013). Note that the pass-through formula (11) requires $\theta < 2$ for $\rho \geq 0$. This is not an additional assumption but rather the second-order condition for the landlord's maximization problem, which must hold for the optimal rent R^* to

be a well-defined interior maximum (See Appendix, Section II.A for derivation).

The parameter θ determines the distribution of the burden for a given $p_e \in (0, 1)$. The pass-through rate is increasing the θ for any given p_e. Equation 11 also allows for the possibility of over-shifting if $\theta > 1$ (strongly log-convex demand) and p_e is sufficiently small. If we assume a simple, linear demand, $q(R) = q_0 - q_1 R$, we have $q'' = 0$ and therefore $\theta = 0$. The pass-through rate is $\rho = (1 - p_e)/2$. Under thus special case, the burden of property taxes is equally shared between the landlord and the tenant if $p_e = 0$, the case in which there is no enforcement on vacant units. In this case, the landlord bears more than half the burden if $\theta < 0$ (log-concave demand), and the tenant bears more than half if $\theta > 0$ (log-convex demand).

VI. EFFECTS ON BUSINESS ACTIVITY

VI.A. Revenues, Profits, and Investment

Property tax liabilities raise occupancy-related costs, directly for owner-occupiers and indirectly through rent pass-through for tenants businesses as highlighted in the previous section. These higher costs may trigger adjustments that affect firms' revenues, profits, investment, and other components of business activity. To examine this, I use a balanced panel of active firms.[10]

Figure 5 presents the effects on log revenues, profits (normalized by pre-reform turnover), and investment. Prior to the reform, there are no differential trends in these outcomes among firms located in treated areas relative to those in untreated areas, lending further support to the parallel trends assumption underlying the identification strategy. Following the reform, firms in treated areas experience a short-term increase in revenues relative to those in untreated areas; however, this difference quickly dissipates. Although I am not able to directly observe prices, this short-lived increase in revenue likely reflects firms' attempt to pass the burden to consumers through higher product prices. If demand is relatively inelastic in the short term, such price adjustments could temporarily raise revenues. Since this effect quickly disappears, it suggests that firms are not able to sustainably shift the burden to consumers. However, Panel B and C show, a differential decline in the profits and investment of firms in treated relative to untreated locations, following the reform. On average, firms' profits declined by 3.3 percent relative to pre-reform revenues, while investment reduced by 2.7 percentage points. Appendix Figure A.3 shows the effects on firms' value added, capital stock, and investment at extensive margin, revealing a dif-

[10] I restrict active firms to those that report non-zero turnover in the data.

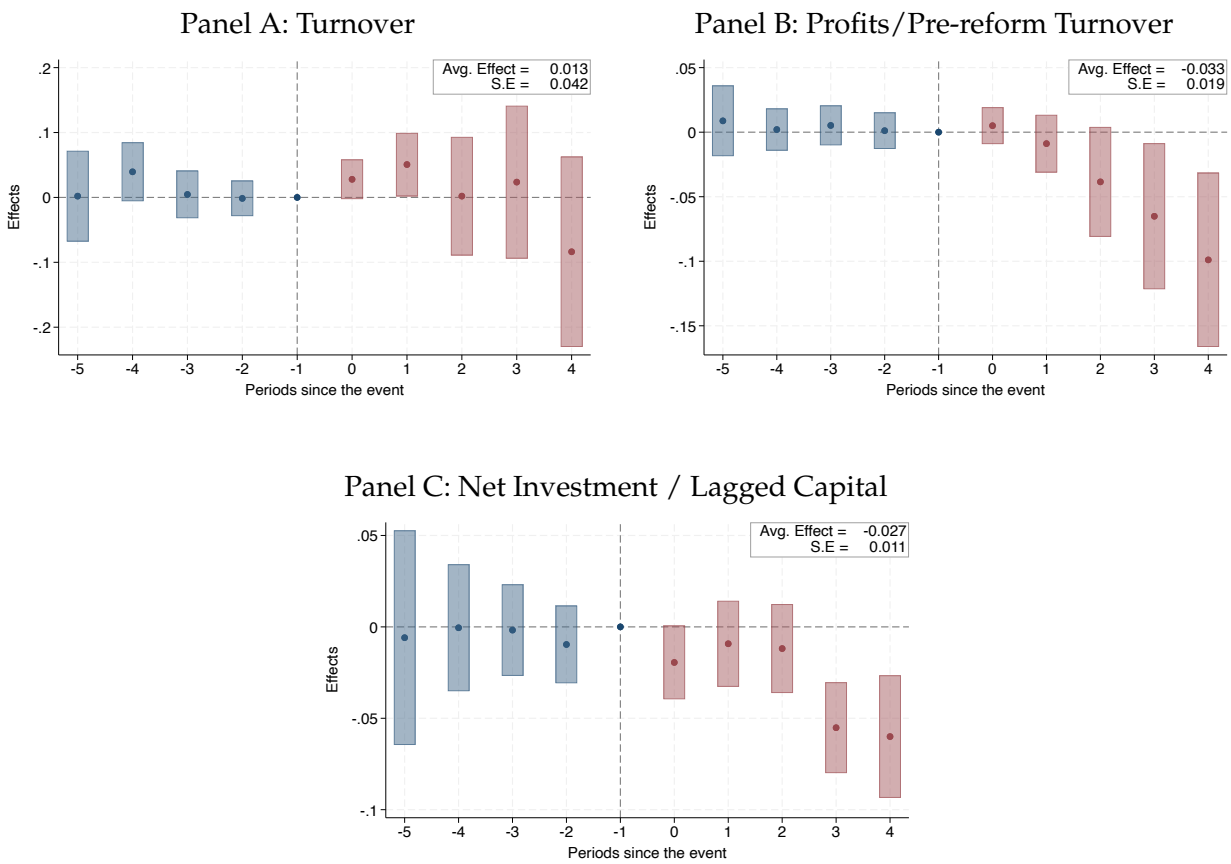


Figure 5: Effects of the Reform on Revenues, Profits, and Investment

Notes: The Figure presents the dynamic effects of the reform on revenues, profits, and investment. Revenues are transformed in logs. Profits are defined as earnings before interest and taxes (EBIT), scaled by pre-reform revenues. Investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

ferential decline in all these outcomes.[11] Average effects are reported in Appendix Table A.1.

VI.B. Firm Survival

The analysis in the previous section is based on a restricted sample of firms that continue operating throughout the sample period. If increases in property tax liabilities drive

[11] The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets.

some firms out of business, the absence of significant effects in the previous section may simply reflect the exit of weaker firms, leaving behind firms that are better able to shift the burden to other margins. In this section, I investigate whether increases in property tax liabilities affect the probability of firm survival.

Constructing the Survival Outcome:—Analyzing firm survival in firm-level panel data is challenging because firm exits inherently renders the sample endogenous to the reform. To address this, I fix the cohort to firms present in 2015/16 (the year prior to the reform), with positive turnover, regardless of whether they operated in other years.[12] I then construct a balanced panel by expanding each firm to all sample years and assigning zeros for turnover, sales, and related variables in the years the firm did not operate. The outcome variable is then defined as an indicator equal to 1 if a firm has positive turnover in year t and 0 otherwise. A similar approach was used by Saez et al. (2019) when analyzing whether payroll taxes affects firm survival in Sweden. By construction, the mean of the outcome variable is one in the reference year; it can be below one before 2016 because some firms had not yet entered, and below one after 2016 because some firms exit following the reform.

However, as highlighted by Saez et al. (2019), using this approach to estimate the effects on firm survival requires reweighting firms to align pre-reform distributions of key variables across different groups, since unweighted estimates can easily reflect compositional differences rather than the causal effects. For example, if younger firms, which tend have lower survival rates, are unevenly distributed across locations that received the reform or not, unweighted estimates may simply reflect age-related compositional differences. Therefore, to analyze compellingly whether property taxes affects firm survival, I reweight the firms by their pre-reform turnover, using the nonparametric reweighting method developed in DiNardo et al. (1996) (DFL reweighting).

Effects on Firm Survival:—Figure 6 shows the effects of the reform on firm survival using the reweighted sample constructed following the procedure described above. In the pre-reform period, there is no evidence of differential trends in survival probabilities between firms located in areas that experienced increases in property tax liabilities and those in areas that did not, as all pre-reform coefficients are statistically indistinguishable from zero. This finding supports the underlying identifying assumption that, absent the reform, survival probabilities for treated and untreated firms would have evolved similarly, allowing post-reform effects to be interpreted as causal effects of the reform.

Following the reform, there is a differential decline in survival probabilities for firms

[12] As a robustness check, I repeat the this procedure using the cohort of firms present in the first year of the sample (with positive turnover)

located in areas that experienced an increase in property tax liabilities relative to firms in locations that did not experience the reform, implying the reform increased firm exit. The negative effects start to manifest after one year of the reform, grows in the second year before it stabilizes at proximately 5 percentage points and persists up to the end of the sample. The corresponding average effects are presented in Table 3. Column 4 shows the preferred specification which includes a full set of controls. Comparing firms in locations that experienced increase in property tax liability versus those in other locations, the reform led to a 3.2 percent decline in survival probabilities with standard error of 1.1 percent.

Based on the total number of active firms in the city in the pre-reform period, this reduced-form estimate corresponds to approximately 443 additional firm exits citywide. Scaling this effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 27 additional firm exits (see bottom row of Table 3). Given that only about 65 percent of potential commercial property tax liabilities are ultimately collected, this estimate further implies that a $1 million increase in collected citywide property tax revenue is associated with approximately 42 additional firm exits citywide.

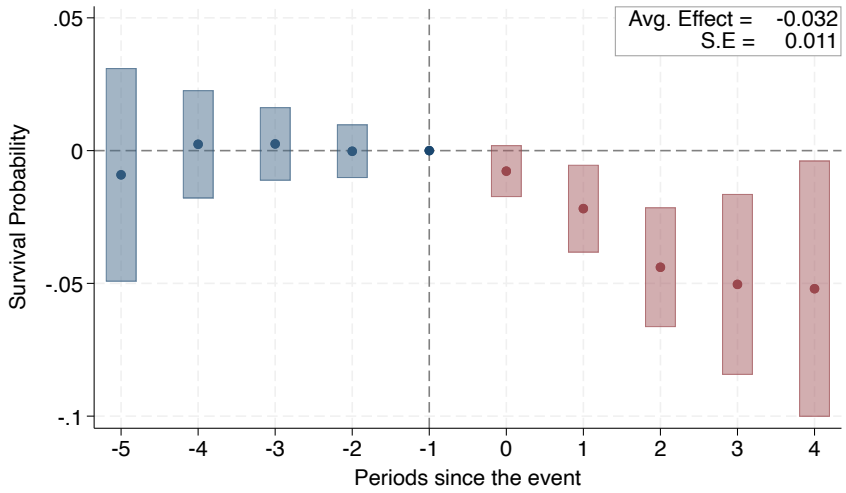


Figure 6: Effects of Property Tax Enforcement on Firm Survival

Notes: The figure presents the dynamic effects of the reform on firm survival. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

Table 3: Effects of Property Tax Enforcement on Firm Survival

	(1)	(2)	(3)	(4)
ATE	-0.036***	-0.029***	-0.029**	-0.032***
	(0.011)	(0.011)	(0.011)	(0.011)
Sector of Operation	No	Yes	Yes	Yes
Pre-reform Size	No	No	Yes	Yes
Other Controls	No	No	No	Yes
Observations	130864	123864	123864	123864
Exits per $1 Million	30.00	24.23	24.03	26.82

Note: The table presents average effects of the reform on firm survival probabilities. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise. Because the mean of the dependent variable is one in pre-reform year by construction, the coefficient estimates can be interpreted as percent change in survival probability. Each column reports estimates from a different specification with an alternative set of controls. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row reports the citywide reform-induced firm exits per additional $1 million in aggregate property tax liability, calculated as $\frac{\hat{\beta} \cdot N}{\Delta T^{\text{City}}}$, where N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide property tax liability. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

A key concern with this approach to measuring firm survival is that it may be confounded by evasion responses. Firms may respond to the higher costs associated with increased property tax liabilities by ceasing to file income tax returns, in which case this approach would misclassify them as exits. I implement several strategies to deal with this concern. First, I use a cross-tax validation strategy to identify firms' operational footprints. Using employee-level tax return data in combination with CIT data, I classify evasion as cases where the firm disappears from the CIT but still has employees associated with its tax identification. Accordingly, I classify a case as exit if the firm disappears from the CIT and has no footprints in the employee tax returns, i.e., both taxes go silent. Appendix Figure A.4 shows that my estimates are unlikely to be driven by an evasion response. A likely criticism of this validation approach is that firms may collude with workers to evade both taxes. However, the timing mismatch between the two tax returns makes such collusion practically difficult. Employee tax returns are filed monthly, whereas corporate income tax (CIT) is filed annually. Since I classify a case as evasion even if the firm's identifier appears in only one month, for collusion to occur, the firm would have to decide to evade at the start of the income year, that is, before the realization of output. Moreover, workers would need to be compensated for agreeing to collude, as doing so would entail forfeiting other benefits such as social security. This

creates an additional constraint on the firm's ability to engage in coordinated evasion. To my knowledge, this paper is the first to cleanly separate survival effects from evasion response using the cross-tax referencing.

Other Robustness Checks:— I conduct several robustness checks to assess the sensitivity of the survival effects, with results reported in the Appendix. First, I redefine exit as the absence of a corporate income tax (CIT) return for at least two consecutive years, thereby excluding firms that temporarily disappear for one year and subsequently re-enter. Second, I impose a stricter definition of exit that requires no CIT return in year t and no record in employee tax returns in both years t and $t+1$. As shown in Figure A.5, the estimated effects remain stable under both definitions. Third, I reconstruct the survival outcome following the same procedure as before but restrict the sample to the cohort of firms that were present and reported positive turnover in the first year of the sample. Figure A.6 shows that the results are consistent with those obtained using the baseline sample. As a final robustness check, I assess whether the findings are sensitive to the choice of comparison group in the Callaway and Sant'Anna (2021) difference-in-differences design. Figure A.7 shows that the results remain robust when using never-treated firms as the comparison group rather than not-yet-treated firms.

VI.C. Labor Market Outcomes

Effects on Formal Employment:— The preceding results on firm survival have direct implications for employment. Firm exits lead to job losses, and surviving firms may also adjust employment in response to the reform. I begin by analyzing employment effects using the broader sample from section VI.B, which includes exiting firms, coding employment as zero following exit. I also examine whether continuously active firms adjust employment in response to the higher costs associated with increased property tax liabilities. For this part, I focus on a balanced panel of firms that remain in operation; that is, firms with positive turnover in every year of the sample.

Because I observe only formal employment, which can be zero for some firms even prior to exit, and because employment is coded as zero after firm exit, I do not transform the outcome using logarithms, as this would complicate the interpretation of coefficients as percentage effects, as highlighted by Chen and Roth (2024). Instead, I normalize employment relative to its pre-reform level (2015/16), setting the normalized outcome to zero when employment is zero in both year t and the pre-reform year. I then winsorize the normalized outcome at the 99th percentile. Normalizing outcomes by pre-reform values is common in the literature (e.g., Saez et al. 2019; Harju et al. 2022) and facilitates an economically meaningful interpretation of the regression coefficients in percent changes

or percentage points. As a robustness check, I also estimate the effects using log-like transformations, i.e., log (1+employment), inverse hyperbolic sine transformation, and in levels. Appendix Figure A.8 show that event-study plot and average effects from this analysis are consistent with those obtained from normalized outcomes.

Figure 7 presents the event-study estimates and the corresponding 95% confidence intervals for the impact of the reform on formal employment, estimated following Callaway and Sant'Anna (2021). Prior to the reform, formal employment in firms located in treated areas follows similar trends to those in untreated areas. Following the reform, employment declines differentially among firms in treated areas relative to those in untreated areas. The negative effect emerges in the reform year, grows for the next three years before it stabilizes at proximately 7 percent and persists up to the end of the sample.

The average effects of the reform on employment over the sample period are presented in Table 4. On average, employment in firms located in treated areas declined by 4.5 percent relative to firms located in untreated areas (see Column 4). Appendix Figure A.9 shows that employment also declines when the sample is restricted to firms active in all years of the sample. This indicates that the overall reduction in employment is not driven solely by firm exits but also reflects within-firm adjustments among surviving firms.

To assess the economic magnitude of these effects, I translate the estimated percent decline in employment into an aggregate citywide measure. Because, I do not observe firm-specific property tax liabilities for reasons highlighted in section V.A, I can not compute firm-level employment elasticities with respect to property tax liabilities. Instead, I aggregate the reduced form employment effects across firms and scale them by the aggregate increase in potential property tax liabilities induced by the reform. The reduced form estimate therefore implies that a $1 million increase in aggregate property tax liabilities is associated with approximately 206 fewer jobs citywide (see bottom row, Table 4, column 4). Further scaling by the share of commercial property tax liabilities that is ultimately collected implies that a $1 million increase in collected citywide commercial property tax revenue is associated with approximately 317 fewer jobs.

While not directly comparable due to differences in the type taxes studied, the negative effects on employment here are consistent with previous studies on payroll taxes (Saez et al. 2019; Egebark and Kaunitz 2018) and corporate taxes (Ljungqvist and Smolyansky 2014; Giroud and Rauh 2019).

Effects on Firms' Wage Bill:— Figure 8 presents the effects on firms' total wage bill and informal wages. I define the informal wage bill as the difference between the wage bill reported on the corporate income tax return and the total wage bill derived from employee-level returns, normalized by the CIT wage bill so that coefficients can be interpreted as

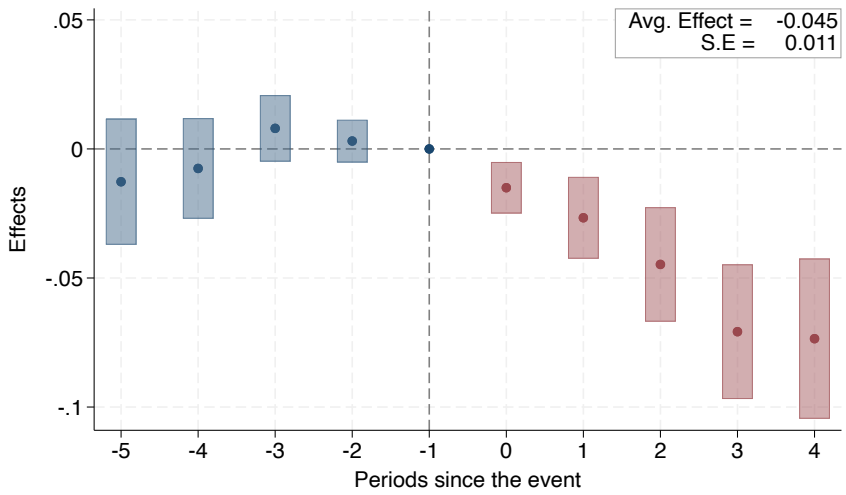


Figure 7: Effects of Property Tax Enforcement on Formal Employment

Notes: The figure presents the dynamic effects of the reform on employment. The dependent variable is normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size. Standard errors are clustered at the firm level.

Table 4: Effects of Property Tax Enforcement on Formal Employment

	(1)	(2)	(3)
ATE	-0.051***	-0.049***	-0.045***
	(0.010)	(0.012)	(0.011)
Industry-Size	N	Y	Y
Other Controls	N	N	Y
Pre-reform Mean	7.429	7.429	7.429
Observations	123921	118584	118584
Jobs Lost per $1 Million	234	225	206

Note: The table presents the average effects of property taxes on the employment, normalized by its pre-reform value. Each column reports estimates from a different specification with an alternative set of control. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row reports the citywide reform-induced job loss per additional $1 million in aggregate commercial property tax liability, calculated as $\frac{\hat{\beta}\cdot\bar{E}\cdot N}{\Delta T^{\text{City}}}$, where $\hat{\beta}$ is the reduced form coefficient, $\bar{E}$ denotes average pre-form employment, N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide commercial property tax liability.

percentage-point changes. Panel A shows a differential decline in the total wage bill of treated firms relative to untreated firms following the reform, with an average effect of 5.5 percent. This effect reflects both the mechanical reduction from the decline in formal

employment documented above and wage adjustments by firms that do not reduce employment. Panel B shows that informal wages increase by 3.3 percentage points from a pre-reform mean of 10.4 percent, suggesting that part of the observed decline in formal employment reflects jobs transitioning to informal arrangements rather than outright job destruction. This partly accounts for the large estimated employment effects.

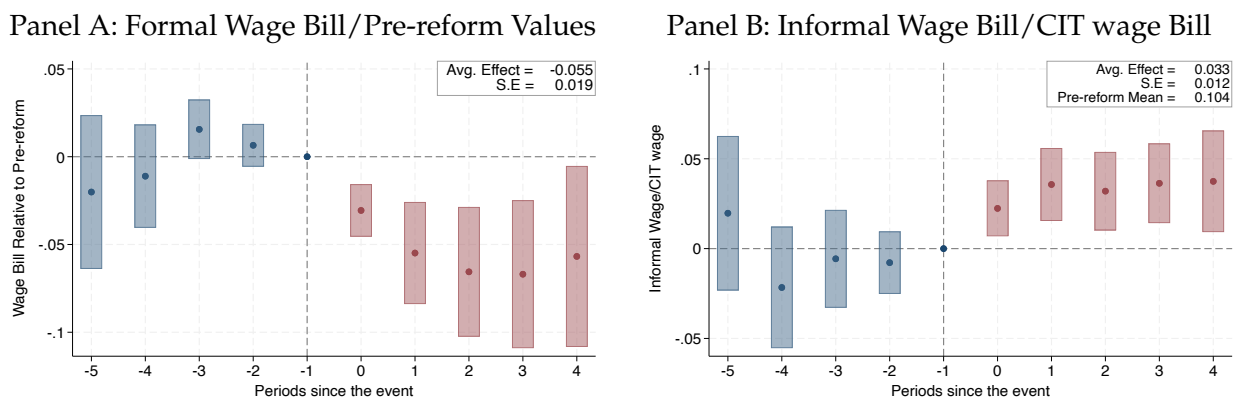


Figure 8: Property Taxes and Firms' Wage Bill

Notes: The figure presents the dynamic effects of the reform on firms' total wage bill. In Panel A, the dependent variable is scaled by its pre-reform value, i.e., $\frac{W_{i,t}}{W_{i,0}}$, where $W_{i,t}$ denotes firm i's formal wage bill at time t and $W_{i,0}$ denotes the pre-reform period wage bill. In Panel B, the dependent variable is informal wage, defined as $\frac{\text{CIT wage bill - Total age bill from employee returns}}{\text{CIT wage bill}}$. This means coefficients can be interpreted as a percent changes. The horizontal axis reports the number of years relative to the policy implementation. Each dot represents the estimated treatment effect for a given year, with year −1 as the reference category. The vertical bands denote the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

VI.D. Heterogeneity Analysis on Firm Survival and Employment

To identify which firms are most vulnerable to increases in property tax liabilities, I estimate heterogeneous treatment effects on firm survival and employment across multiple dimensions. Specifically, I examine heterogeneity by occupancy status (tenant versus owner-occupier), firm size (measured by the number of formal employees), profitability, and leverage. Examining this heterogeneity also sheds light on the mechanisms through which property taxes affect firm outcomes, as well as the distributional consequences of the reform.

Figure 9 summarizes the heterogeneity results on both firm survival and employment. For each dimension, I also estimate event-study coefficients to assess pre-trends within each category. The resulting event-study plots are reported in the Appendix: Figures

A.10, A.11, A.12, and A.13 present heterogeneous effects on firm survival, while Figures A.14, A.15, A.16, and A.17 report the corresponding employment effects.

I begin by examining whether the effect on firm survival varies by firms' occupancy status—specifically, whether the firm operates as a tenant or owns land and buildings. This dimension of heterogeneity is motivated by the "first-stage" results in Section V.A, which show that the reform led to an increase in business rental expenses with substantial pass-through. I classify firms as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Because some firms report both rent expenses and building values, there is overlap between these categories. For firms reporting neither, I assign occupancy status based on pre-reform capital intensity (the ratio of fixed assets to total assets), classifying firms below the median as tenants and those above the median as owner-occupiers. As a robustness check, I show that excluding firms with imputed occupancy status yields qualitatively similar results, albeit with reduced statistical precision.

Panel A of Figure 9 shows that while the reform reduced formal employment for both tenant and owner-occupier firms, the effect is statistically significant only among tenant businesses. Similarly, Panel B indicates that the negative effects on firm survival probabilities are concentrated among tenant firms. This pattern reinforces the pass-through evidence in Section V.A and further contrasts with the predictions of neoclassical theory. The absence of statistically significant effects on either of these outcomes among firms that report owning land and buildings may reflect scale effects: such firms are likely larger and better able to absorb higher property tax liabilities by adjusting other margins. Consistent with this interpretation, heterogeneity by firm size points in the same direction.[13] Negative survival effects are concentrated among the smallest firms, those with fewer than five workers. Employment effects are significant only for firms with fewer than five workers and for those employing five to nineteen workers, with no significant effects among larger firms employing twenty or more workers.[14]

To shed light on potential mechanisms, I examine heterogeneity by baseline profitability and leverage. Property taxes raise fixed costs and thereby reduce profitability; firms with thin operating margins may be pushed into losses, increasing their likelihood of exit. To assess this channel, I first examine effects among loss making firms. I then partition the sample into terciles and examine how the effects vary across the distribution

[13] The firm size categories follow the World Bank's classification for firms in developing countries.

[14] Although firms employing five to nineteen workers exhibit significant employment effects, the event-study estimates in Appendix Figure A.15 show evidence of pre-trends for this group, complicating a causal interpretation of the average effect. By contrast, there is no evidence of pre-trends for firms with fewer than five workers or for those employing twenty or more workers.

of pre-reform profitability.[15] Figures 9 shows that the magnitude of both employment and survival effects declines monotonically with profitability. The event-study plots in Appendix Figures A.12 and A.16 show that this pattern is not driven by pre-trends. The finding that survival effects are strongest among loss-making firms further rules out the possibility that the overall survival effects are driven by evasion responses to corporate income tax, since such firms have zero corporate income tax liability.

Turning to leverage, higher property tax liabilities increase operating costs and tighten already thin repayment margins for highly leveraged firms, thereby amplifying financial distress. Figure 9 shows that negative employment and survival effects are only significant among firms with very low and very high leverage, but not those with moderate leverage. This pattern of results suggest that moderate leverage may help firms buffer cost shocks by allowing them to draw on relationships with lenders as a coping strategy. By contrast, very high leverage appears to exacerbate financial distress in the presence of cost shocks.

Taken together, the heterogeneity patterns by profitability and leverage point to financial constraints and financial distress as central mechanisms underlying the reform's effects on firm survival and employment.

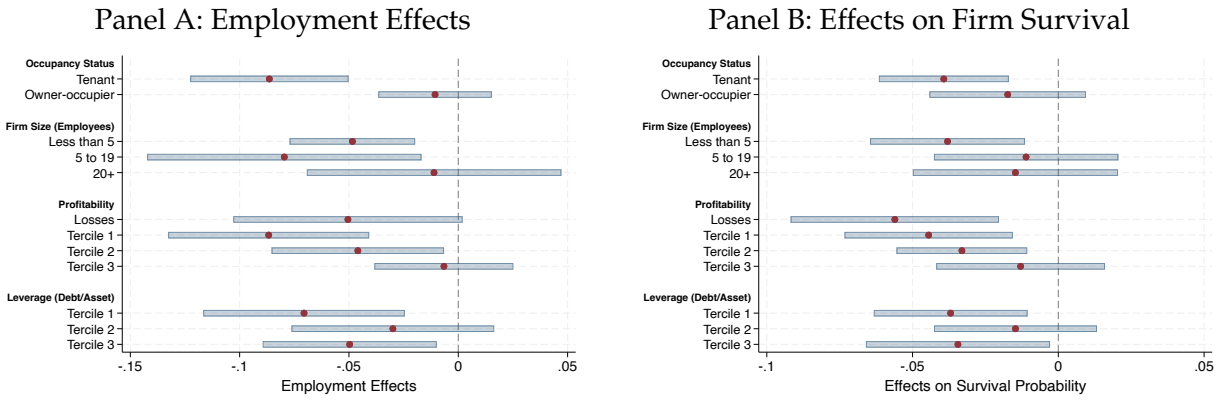


Figure 9: Employment and Survival Effects: Heterogeneity by Firm Characteristics

Notes: The figure shows the heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient from a separate regression. Firms are distinguished by pre-reform characteristics based on (i) occupancy status(tenant business or owner-occupier); (ii) firm size (number of formal employees); (iii) profitability (EBIT/revnues); and (iv) leverage (debt-to-asset ratio). The regression specifications includes fixed effects for industry, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

[15] Profit margins are defined as the ratio of profits (EBIT) to revenue.

VII. EFFECTS ON WAGES AND DISTRIBUTIONAL CONSEQUENCES

The aggregate employment effects mask important distributional consequences, as different workers may be affected differently. Moreover, firms may pass the higher costs associated with property taxes on to workers through wage adjustments without necessarily reducing employment. As with other taxes, such as business owner taxes (e.g., Risch 2024), the burden may not be evenly distributed across workers.

To identify which types of workers are most affected by the reform, I pursue two complementary analyses. First, I examine the effects of the reform on firms' wage distributions by estimating how the within-firm 25th, 50th, and 75th percentiles of wages evolve before and after the reform in treated relative to untreated firms.[16] Second, I study changes in the composition of employment by tracking, relative to pre-reform total employment, the shares of workers in each pre-reform wage quartile over time.[17] In both analyses, I restrict the sample to firms that employed at least five workers in the pre-reform period and remained active throughout the sample period to ensure adequate support across the wage distribution.[18] I conduct the analysis using a similar approach as before, i.e., staggered DiD, following Callaway and Sant'Anna (2021).

Figure 10, Panel A, presents the event-study estimates and the corresponding 95 percent confidence intervals for the impact of the reform on the 25th, 50th, and 75th percentiles of wages. The quantiles are scaled by the pre-reform total formal wage bill; accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the wage bill. Prior to the reform, wage quantiles in firms located in treated areas exhibit trends similar to those in untreated areas, supporting the parallel trends assumption. Following the reform, there is a differential decline in the 50th and 75th wage percentiles in firms located in treated areas relative to those in comparison locations, with the decline larger and statistically significant at the 75th percentile.

Panel B of Figure 10 presents the average effects on each wage quantile, expressed in percentage points relative to the pre-reform formal wage bill. While wages decline across all quantiles, the effects are stronger and statistically significant at the top of the distribution. In particular, the 75th percentile of wages declines by 0.18 percentage points relative to the pre-reform wage bill. As a robustness check, I repeat the analysis normalizing wage quantiles by their own pre-reform values rather than by the total wage bill, which allows the coefficients to be interpreted as percent changes. Appendix Figure A.19 shows that

[16] The focus on the 25th, 50th, and 75th percentiles reflects sample size limitations.

[17] I conduct the analysis at the firm level rather than the worker level, as workers are uniquely identified only at the firm level in the data.

[18] Coincidentally, there is no significant effect on employment in this sample (see Appendix Figure A.18)

the 75th percentile of wages declined by 5.6 percent in treated firms relative to untreated firms, while no significant effects are observed at the 25th or 50th percentiles. Further, I analyze the effects on the wage distribution using four quantiles, the 20th, 40th, 60th, and 80th percentiles, as outcome variables. Appendix Figure A.20 shows a pattern identical to that in Figure 10, both for the event-study estimates (Panel A) and the average effects (Panel B), with larger and statistically significant effects at the 60th and 80th percentiles. These results imply that the burden associated with higher property tax liabilities falls disproportionately on higher earning workers within firms, suggesting that the reform reduced within-firm wage inequality between low and high wage workers.

To shed more light on the distributional consequences of the reform, I examine changes in worker composition using the same sample. I construct bins based on the pre-reform wage distribution and track how the share of workers in each bin, relative to pre-reform total employment, evolves before and after the reform. Consistent with the wage analysis above, I use quartiles as the bins. The outcome variable is therefore $\frac{E_{q,t}}{E_0}$, where $E_{b,t}$ denotes the number of workers in quartile q at time t and E_0 denotes pre-reform total employment.

Figure 11 shows that the reform led to a differential decline in the share of workers in the top quartile of the pre-reform wage distribution among firms located in treated areas relative to those in untreated areas. This decline is accompanied by an increase in the share of workers in the 50th–75th percentile. Panel A shows that these patterns are not driven by pre-trends, supporting a causal interpretation. On average, the share of workers in the top quartile declined by 2.8 percentage points in treated firms relative to untreated firms, while the share in the 50th–75th percentile increased by 2.2 percentage points (Panel B). There is no statistically significant average effect on the share of workers in either the bottom quartile or the 25th–50th percentile, although the point estimates are positive.

This pattern of results may reflect two mechanisms. First, wages at the top of the distribution may have declined more than those at the bottom, consistent with the evidence in Figure 10, thereby shifting workers into lower wage bins. Alternatively, firms may substitute higher earning workers with lower wage workers while keeping overall employment constant. Both mechanisms imply that workers at the top of the pre-reform wage distribution are disproportionately affected by the reform. Taken together, the results in Figures 10 and 11 suggest that the reform was somewhat progressive and led to skill downgrading within firms.

These findings are related to Risch (2024), who shows that a large share of the tax burden on business owners is borne by relatively high-earning workers. Even though

property taxes are different from business owner taxes, both taxes are not levied directly on business income.

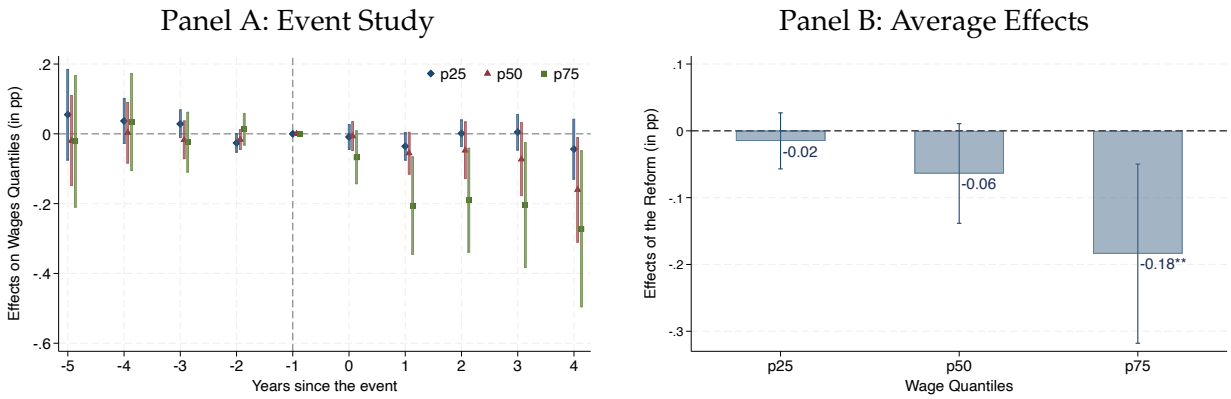


Figure 10: Effects on Wage Quantiles: Percentage Points Relative to Pre-reform Wage Bill

Notes: The figure presents the effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specifications includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

VIII. CONCLUSION

This paper examines the effects of commercial property taxes on real activity and the distributional consequences at firm level. I exploit a natural experiment in Kampala, the capital city of Uganda, where property tax liabilities increased following a revaluation and mass property enumeration exercise, while surrounding areas were unaffected. This setting allows me to estimate causal effects using a difference-in-differences identification strategy. I use data from corporate income tax returns and employee tax returns.

First, I show that the reform increased business rental costs by 9 percent, with an estimated citywide pass-through rate of 0.42, sharply contradicting the neoclassical prediction that landlords should bear the entire burden. To explain this contrast, I propose a framework that embeds a key institutional feature of selective enforcement: the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity, and lower when the unit is unoccupied. Second, I show that while the reform had no significant effects on revenues, it reduced firms'

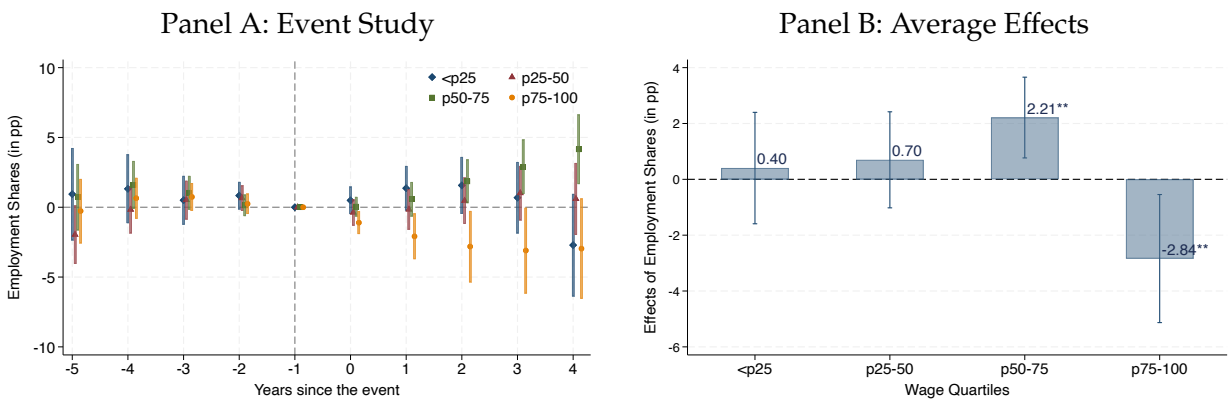


Figure 11: Effects of the Reform on Employment Composition

Notes: The figure presents the effects of the reform on within firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,b,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform total employment. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specifications includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

profitability, investment, employment and survival probabilities.

Finally, I examine the distributional consequences of the reform by assessing which types of workers are most affected. First, I study how within-firm wage quantiles evolve before and after the reform in treated firms relative to untreated firms. Second, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the 75th percentile of wages declines differentially in treated firms relative to untreated firms, while there is no statistically significant effect on the 25th and 50th percentiles, despite negative point estimates. Consistent with this pattern, I also find a decline in the share of workers in the top quartile of the pre-reform wage distribution, alongside an increase in the share of workers in the 50th–75th percentile. Taken together, these results suggest that the reform reduced within-firm wage inequality and was therefore progressive, a pattern consistent with skill downgrading within firms.

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I. APPENDIX: ADDITIONAL FIGURES AND TABLES

I.A. Additional Figures

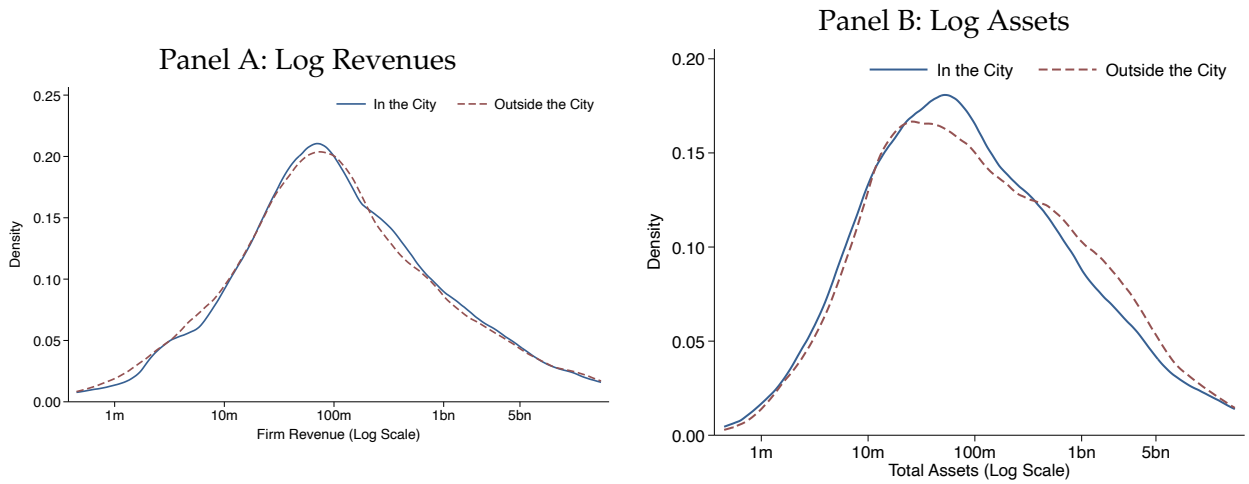

Figure A.1: Firm Size Distributions Across Locations

Notes: This figure plots the size distribution (in log scale) of pre-reform active firms located inside (treated) and outside (comparison) the city. Panels A shows the distribution of revenues while Panel B shows the distribution of assets.

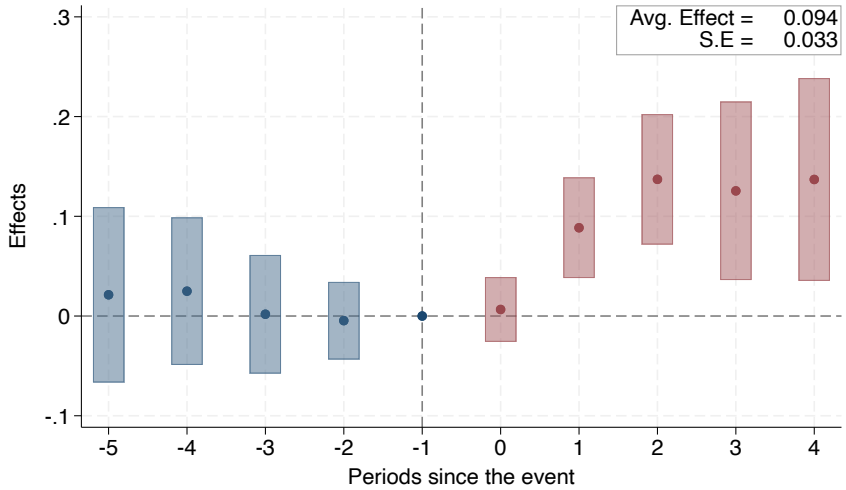

Figure A.2: Effects of Property Taxes on Business Rental Expenses: Never Treated as Comparison Group

Notes: The figure shows the dynamic effects of property taxes on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change after multiplying by 100. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with never treated units used as the comparison group. Standard errors are clustered at the firm level.

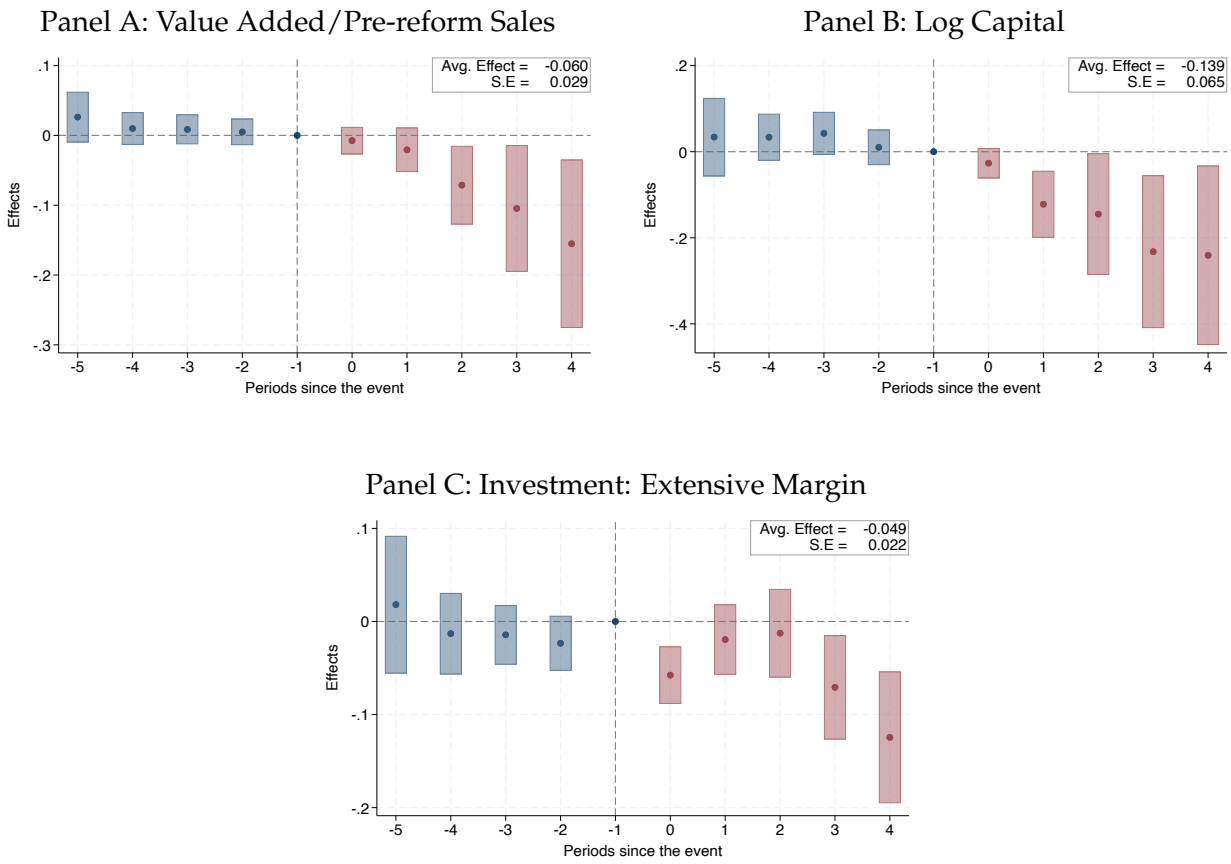

Figure A.3: Effects on Value Added, Capital, and Extensive Margin Investment

Notes: The Figure presents the dynamic effects of the reform on firms' value added, capital (in logs) and investment (extensive margin). Value added is defined as revenues minus cost of sales, normalized by pre-reform sales. Capital is defined as the book value of net fixed assets. Investment is defined as the annual change in fixed assets. At the extensive margin, the outcome is one if the firm had positive investment, zero otherwise. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

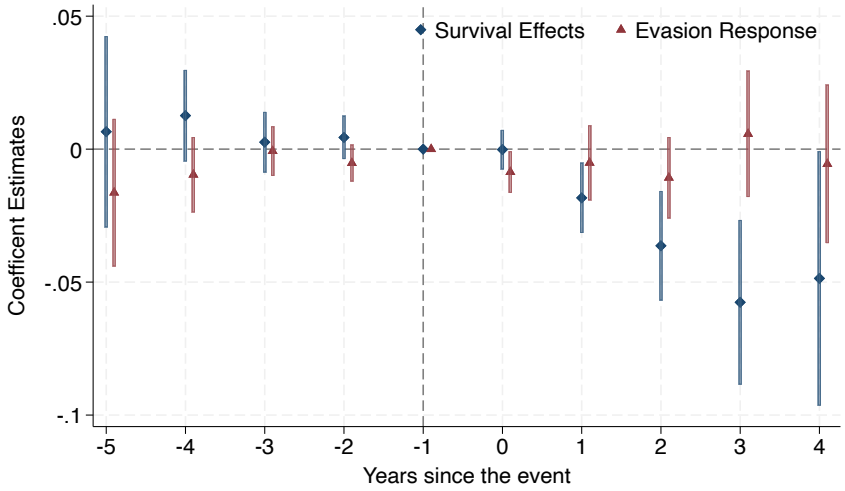

Figure A.4: Effects on Firm Survival vs Evasion

Notes: The figure presents the dynamic effects of the reform on firm survival, separating survival from evasion response. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector and 2016 size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

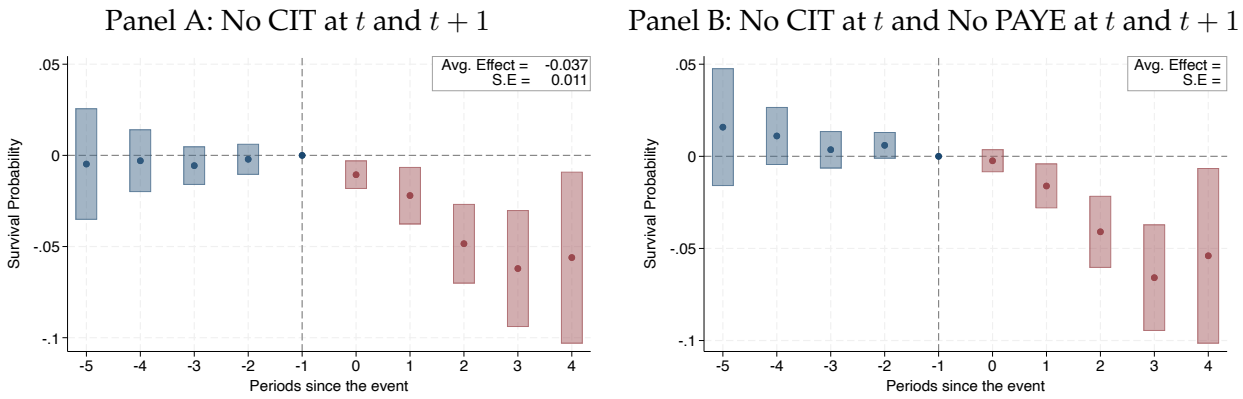

Figure A.5: Property Tax Enforcement and Firm Survival: Robustness at t and $t+1$

Notes: The figure presents the dynamic effects of the reform on firm survival. In Panel A I define exit as the absence of corporate income tax (CIT) return for at least two consecutive year. In Panel B restricts exit to firms that have no CIT return in year t and no footprints in the employee tax returns in both year t and $t+1$. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

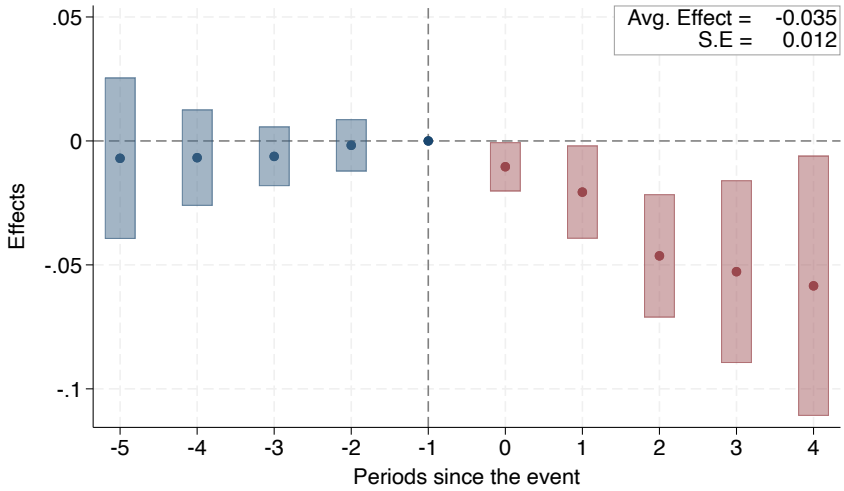

Figure A.6: Property Tax Enforcement and Firm Survival: Sample of Firms Present in First Year

Notes: The figure presents the dynamic effects of the reform on firm survival, with the sample fixed to firms active in the first year of the sample rather than the year immediately prior to the reform. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

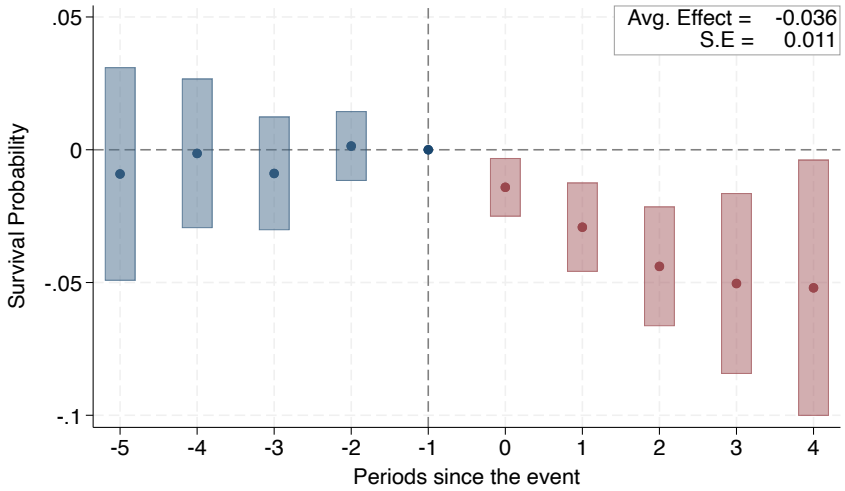

Figure A.7: Effects of Property Tax Enforcement on Firm Survival: Never-Treated as Comparison Group

Notes: The figure presents the dynamic effects of the reform on firm survival, using never treated units as the comparison group rather than not-yet treated. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform, with negative values representing pre-treatment periods. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. Standard errors are clustered at the firm level.

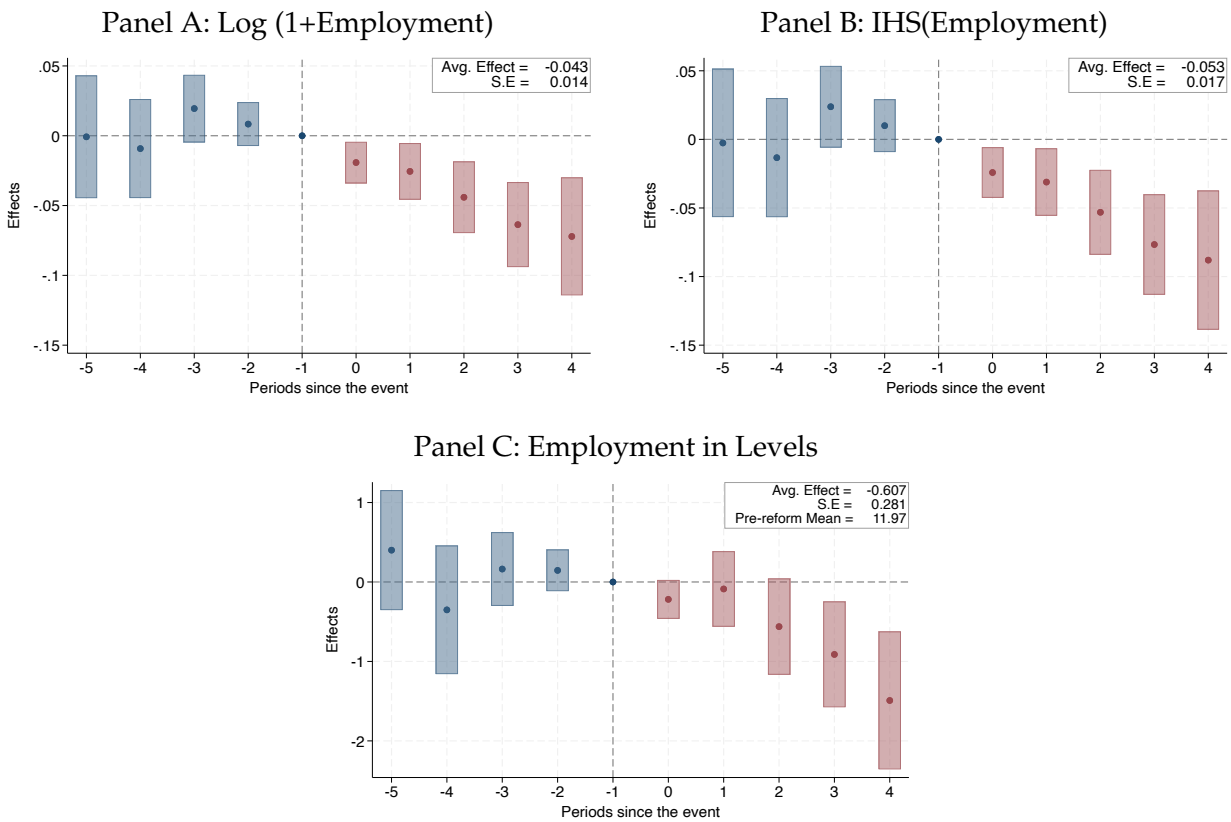

Figure A.8: Effects on Formal Employment: Logs, IHS, and Levels

Notes: The figure presents the dynamic effects of the reform on firms' formal employment. In Panel A employment is transformed using logs, i.e log(1+employment). In Panel B employment is transformed using inverse hyperbolic sine transformation. Panel C presents the effects in levels where the scale on the vertical axis is number of jobs. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size and accounting month. Standard errors are clustered at the firm level.

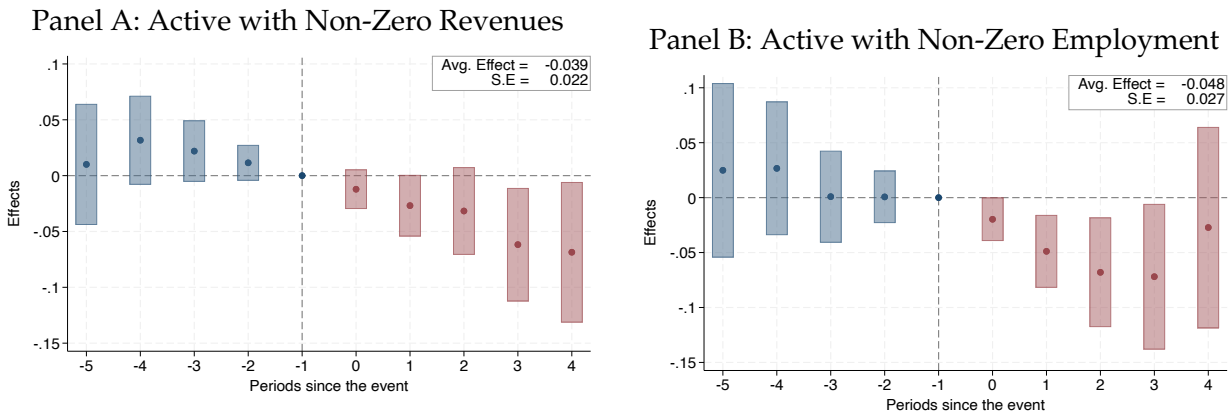

Figure A.9: Effects on Formal Employment: Sample of Firms Active in all Sample Years

Notes: The figure presents the dynamic effects of the reform on formal employment with the sample restricted to a panel of firms that are active in all the years of the sample. Panel A restricts the sample to firms with non-zero revenues in all years, normalizing the dependent variable by its pre-reform value. Panel B restricts the sample to firms with non-zero workers in all years, transforming the outcome variable using logs. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

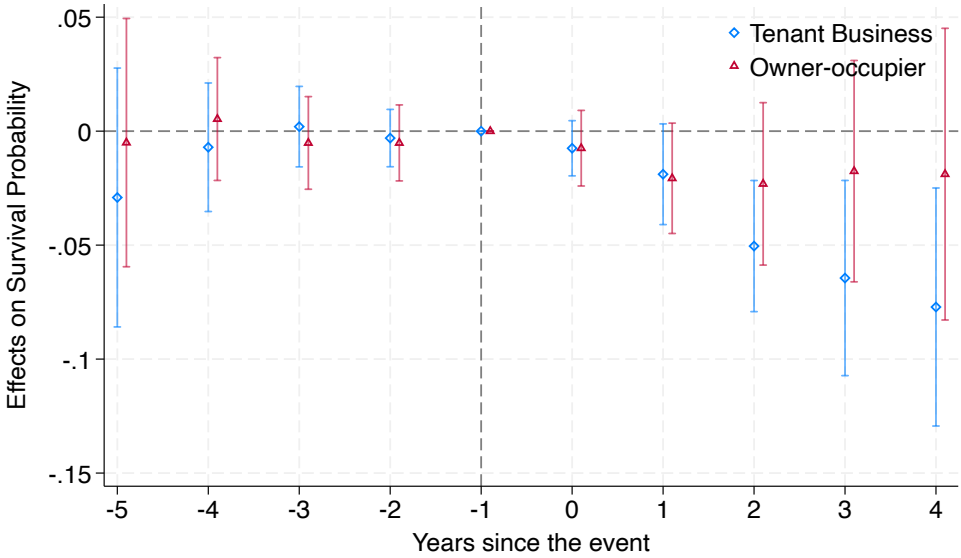

Figure A.10: Survival Effects: Heterogeneity by Occupancy Status

Notes: The figure shows the dynamic effects of the reform on firm survival, separately for tenant and owner-occupier firms. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned ownership status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

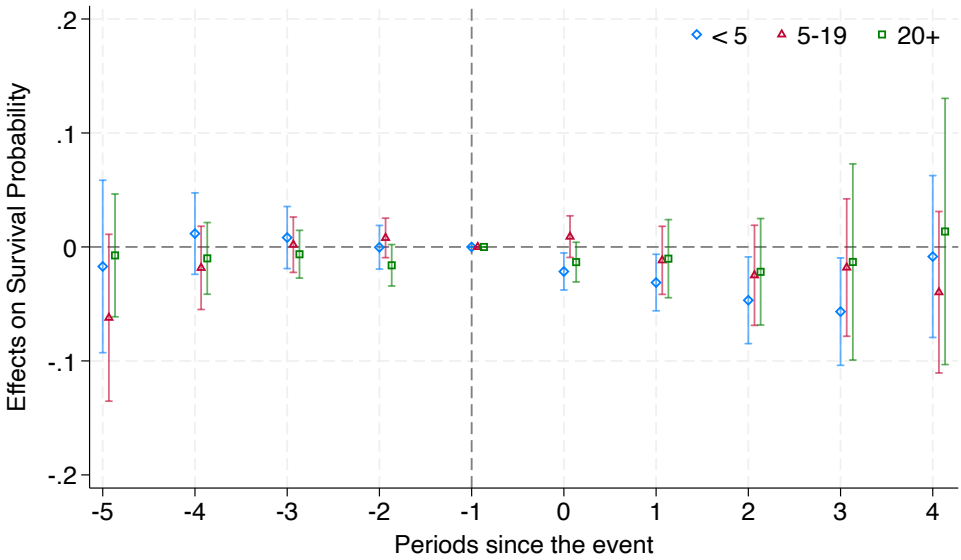

Figure A.11: Survival Effects: Heterogeneity by Size (Number of Formal Employees)

Notes: The figure shows the dynamic effects of the reform on firm survival, separately for different firm sizes based on their pre-reform formal employment. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

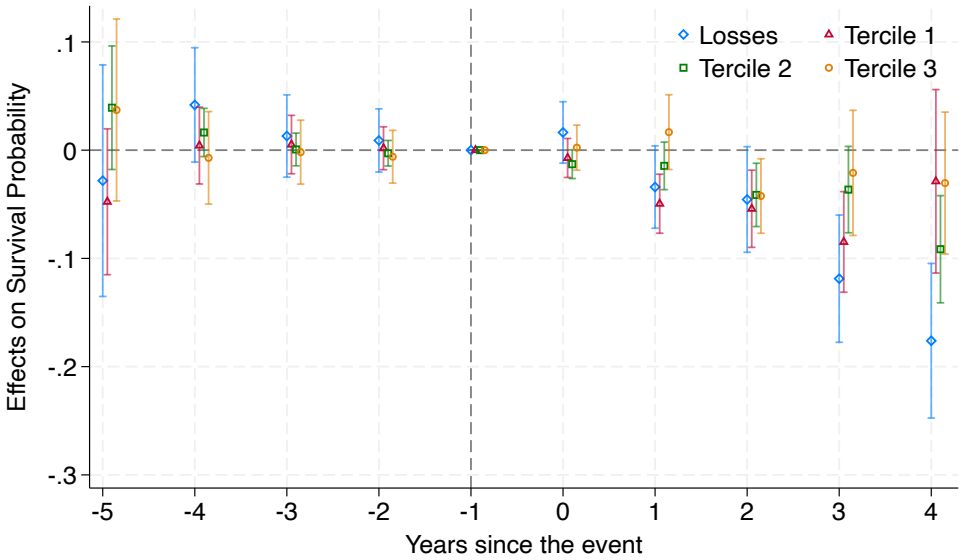

Figure A.12: Survival Effects: Heterogeneity by Profitability

Notes: The figure shows the dynamic effects of the reform on firm survival for firms at different points in the pre-reform profit margins (EBIT/revenues) distribution. The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

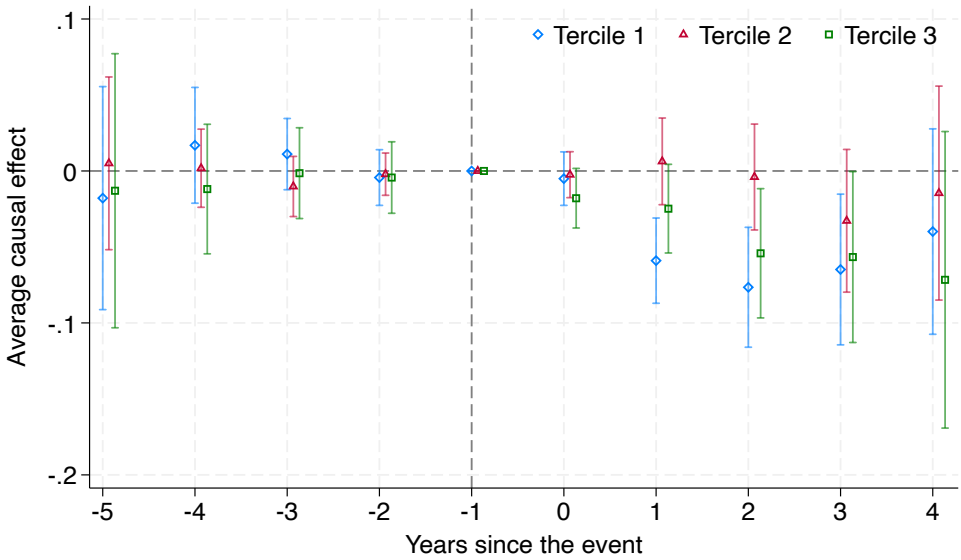

Figure A.13: Survival Effects: Heterogeneity by Level of Leverage

Notes: The figure shows the dynamic effects of the reform on firm survival for firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is an indicator equal to one if the firm is operating at time t, zero otherwise, implying the coefficients can be interpreted as percentage point change in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

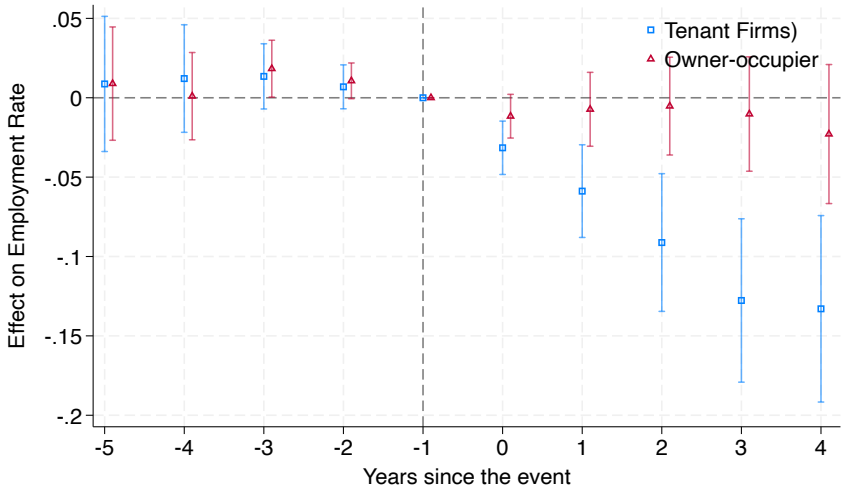

Figure A.14: Employment Effects: Heterogeneity by Occupancy Status

Notes: The figure shows the dynamic effects of the reform on formal employment, separately for tenant and owner-occupier firms. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned ownership status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

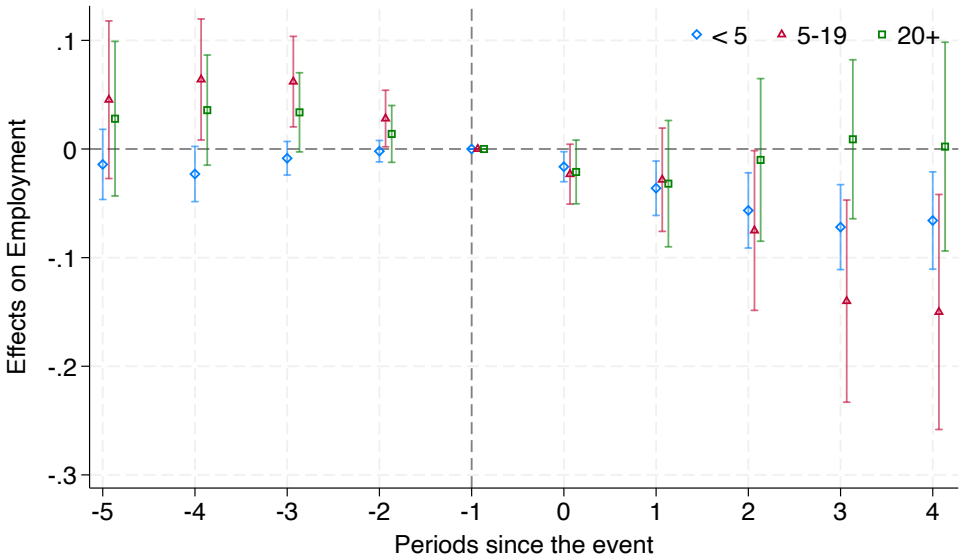

Figure A.15: Employment Effects: Heterogeneity by Size (Number of Formal Employees)

Notes: The figure shows the dynamic effects of the reform on formal employment, separately for different firm sizes based on their pre-reform formal employment. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

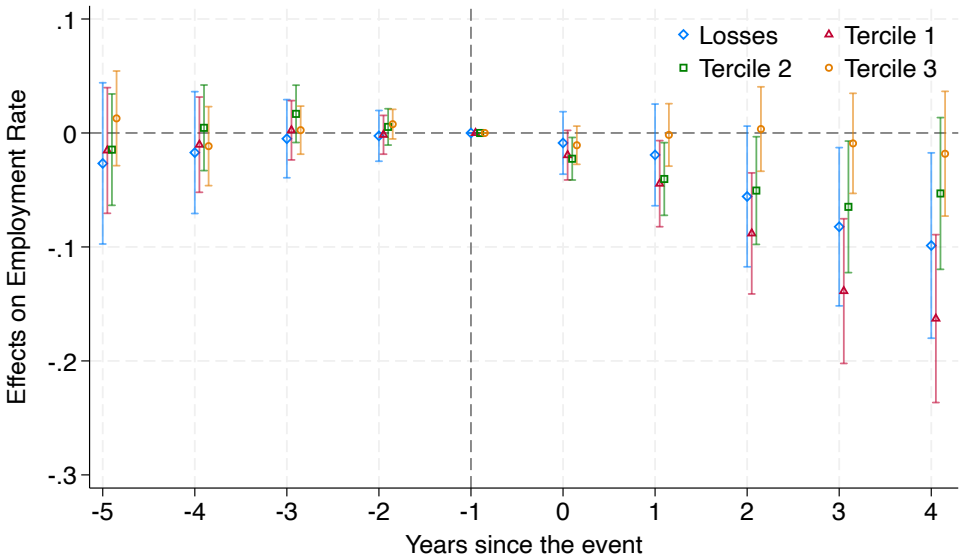

Figure A.16: Employment Effects: Heterogeneity by Profitability

Notes: The figure shows the dynamic effects of the reform on formal employment for firms at different points in the pre-reform profit margins (EBIT/revenues) distribution. The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

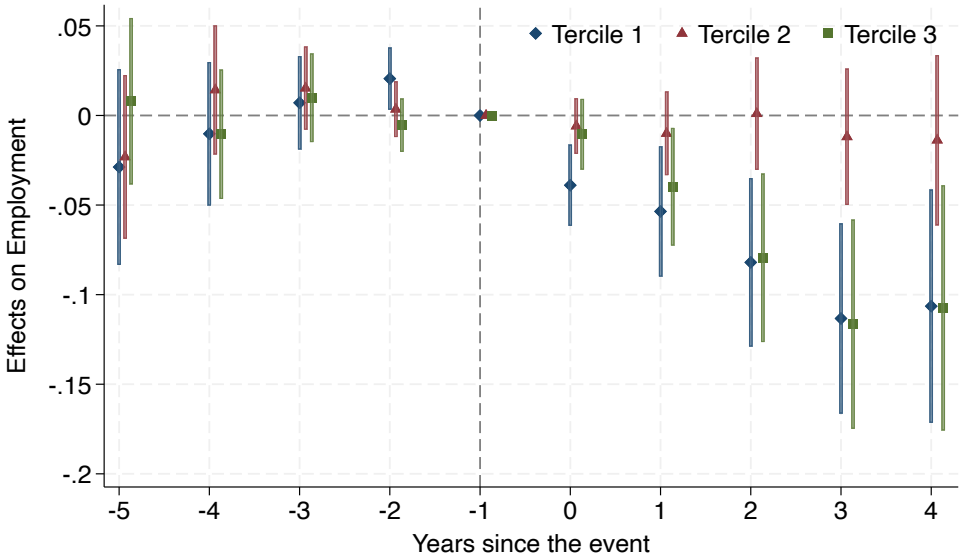

Figure A.17: Employment Effects: Heterogeneity by the Level of Leverage

Notes: The figure shows the dynamic effects of the reform on formal employment for firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is number of workers, normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at −1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach, following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

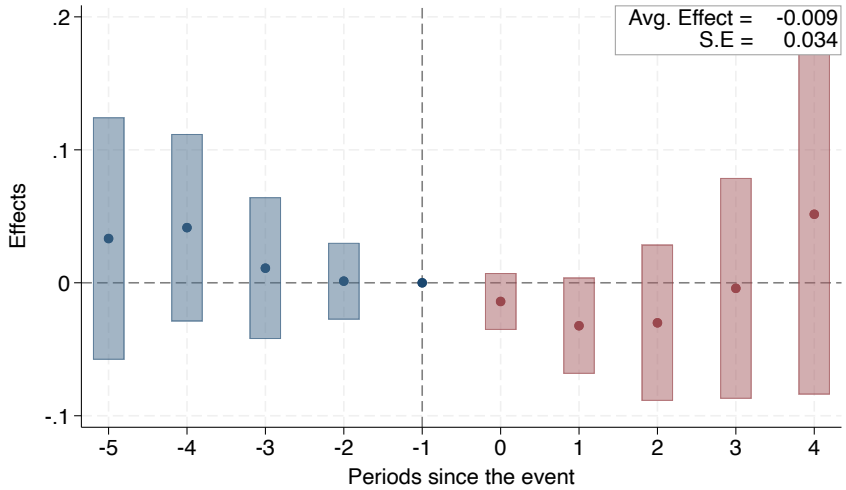

Figure A.18: Effects on Formal Employment: Sample of Active Firms with at least Five Workers in Pre-reform

Notes: The figure presents the employment effects of the reform with the sample restricted to active firms that employed at least five workers in the pre-reform period. The outcome is variable normalized by its pre-reform value, implying the coefficients can be interpreted as a percent changes. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at −1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

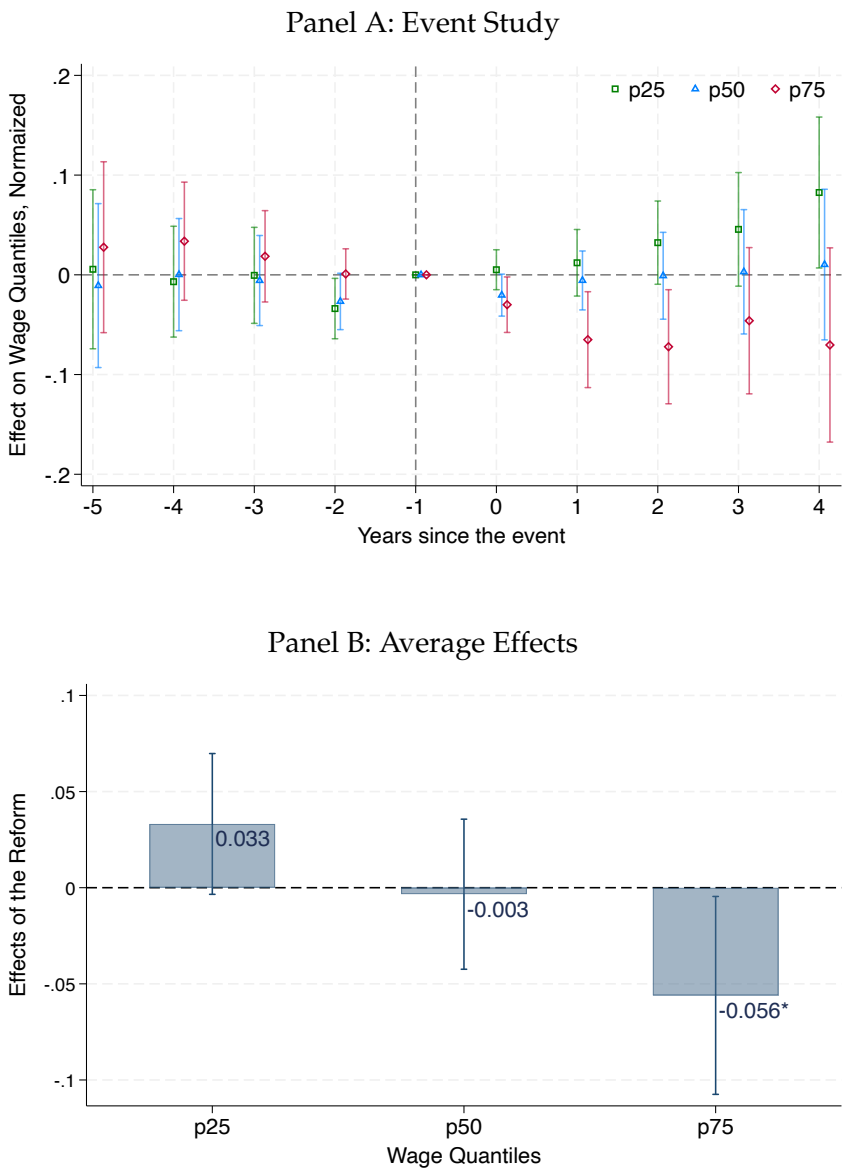

Figure A.19: Effects on Wage Quantiles, Scaled by their Pre-reform Values

Notes: The figure presents the effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th within-firm percentiles of wages, scaled by their pre-reform Values, i.e., $\frac{q_{i,t}}{q_{i,0}}$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $q_{i,0}$ denotes the corresponding quantile in pre-reform period. Accordingly, the coefficient estimates should be interpreted as percent changes. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences design following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

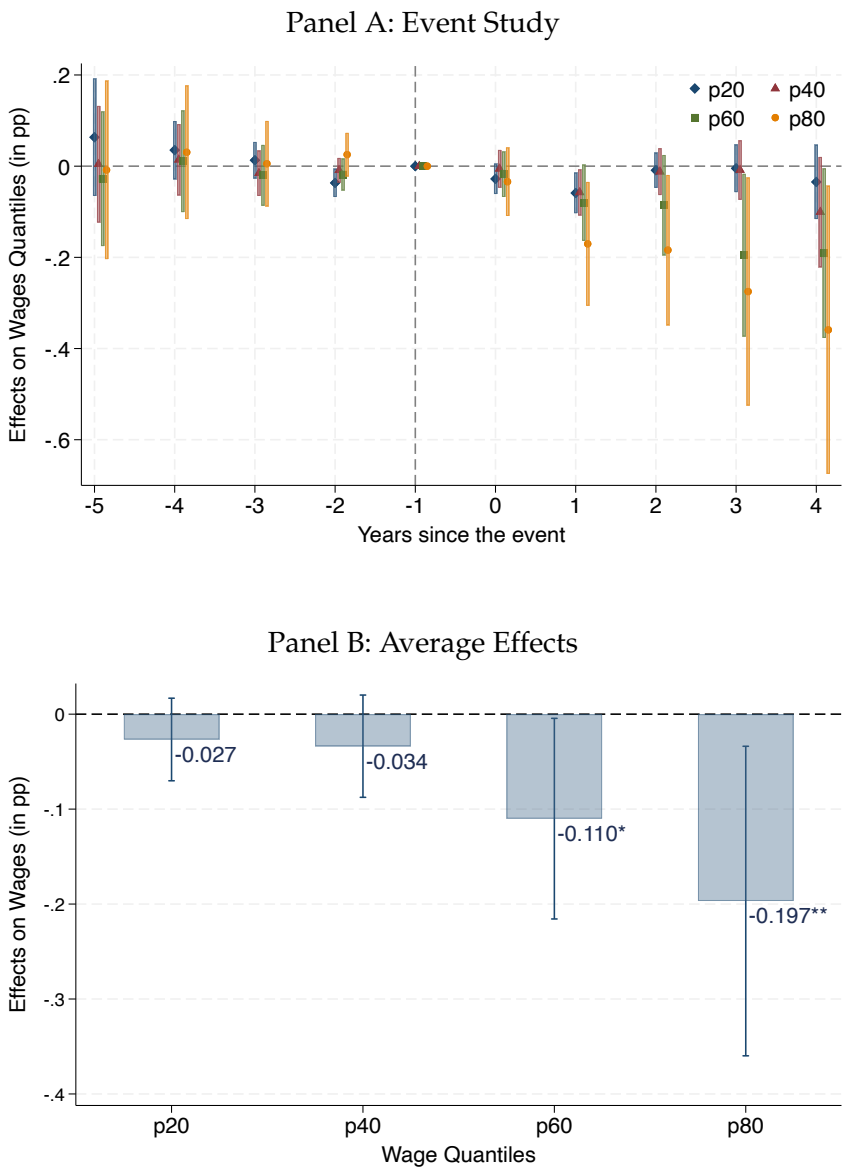

Figure A.20: Effects on Wage Quantiles: 20th, 40th, 60th, and 80th percentiles

Notes: The figure presents the effects of the reform on within-firm wage quantiles. The dependent variables are the 20th, 40th, 60th, and 80th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences design following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

I.B. Additional Tables

Table A.1: Effects of the reform on Firms' Revenues, Profits, Capital, and Investment

	(1) Log Revenues	(2) Profits	(3) Log Capital	(4) Investment/Lagged Capital	(5) Investment: Extensive
ATT	0.013	-0.033*	-0.139**	-0.027**	-0.049**
	(0.042)	(0.019)	(0.065)	(0.011)	(0.022)
Sector of Operation	Y	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y	Y
Observations	33012	33012	29097	43029	43029

Note: The table presents the average effects of the reform on firms' revenues, profits, capital, and investment. Profits are defined as earnings before interest and taxes (EBIT) (sales minus cost of sales, operating and administration expenses). These are scaled by pre-reform revenues. Investment is defined as the annual change in fixed assets, normalized by lagged fixed assets. The last column present extensive margin response of investment. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Each regression includes fixed effects for sector, and pre-reform size. Under "Other Controls," I include dummies for accounting month and whether the firm filed returns on time, all measured in the pre-reform period. Standard errors are clustered at the firm level.

Table A.2: Effects on Wage Distribution: 25th, 25th, and 75th Percentiles

	(1) p25	(2) p50	(3) p75
ATE	-0.015	-0.064	-0.184**
	(0.025)	(0.045)	(0.082)
Sector of Operation	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes
Pre-reform Mean	0.372	0.496	0.816
Observations	19125	19125	19125

Note: The table presents the average effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.3: Effects on Wage Distribution: 20th, 40th, 60th, and 80th Percentiles

	(1) p20	(2) p40	(3) P60	(4) p80
ATE	-0.027	-0.034	-0.110*	-0.197**
	(0.026)	(0.033)	(0.064)	(0.099)
Sector of Operation	Yes	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes
Pre-reform Mean	0.351	0.439	0.583	0.950
Observations	19125	19125	19125	19125

Note: The table presents average effects of the reform on the wage distribution. The dependent variables are the 20th, 40th, 60th, and 80th within-firm percentiles of wages, scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.4: Property Taxes and Worker Composition

	(1) $< p25$	(2) p25-50	(3) p50-75	(4) p75-100
Average Effect (in pp)	0.404	0.698	2.212**	-2.838**
	(1.213)	(1.046)	(0.879)	(1.394)
Sector of Operation	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y
Pre-reform Mean	21.411	22.885	24.831	29.341
Observations	19125	19125	19125	19125

Note: The table presents the average effects of the reform on within firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,b,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the pre-reform total employment. The sample is comprised of active firms with at least five workers in the pre-reform period. All models include fixed effects for sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

II. APPENDIX: MODELS

II.A. Theoretical Framework: Detailed Steps

Landlord's Maximization Problem:— The landlord sets rent to maximize his expected utility:

$$\max_R \; q(R)\,(R-T) + (1-q(R))\,(-p_e T)\,, \tag{12}$$

where $q(R)$ denotes the probability that the unit is filled at rent R, $p_e \leq 1$ is the probability that the tax is enforced given that unit is vacant. The first term the landlord's problem is the expected payoff if the unit is filled: the landlord earns $R-T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1-q(R)$.

Expanding (12):

$$\max_R \; q(R)R - q(R)T - p_e T + q(R)p_e T \tag{13}$$

$$= \max_R \; q(R)R - q(R)(1-p_e)T - p_e T \tag{14}$$

The last term, $-p_e T$, does not depend on R and drops out of the optimization. The landlord effectively maximizes:

$$\max_R \; q(R)\,[R-(1-p_e)T] \tag{15}$$

The landlord behaves as if facing a per-unit cost of $(1-p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero— the landlord pays T regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax T— each occupied unit triggers the entire tax bill.

The first-order condition of (15) is:

$$q'(R)\,[R-(1-p_e)T] + q(R) = 0 \tag{16}$$

Solving for R:

$$\boxed{R^* = (1-p_e)T - \frac{q(R^*)}{q'(R^*)}} \tag{17}$$

The optimal rent has two components. The first, $(1-p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape

of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

Characterizing the Pass-through Rate:— Defining $c \equiv (1 - p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent R^* as a function of the effective cost c, and hence of T. Define the function:

$$G(R, c) \equiv q'(R)(R - c) + q(R) = 0 \tag{18}$$

By the implicit function theorem:

$$\frac{dR^*}{dc} = -\frac{\partial G/\partial c}{\partial G/\partial R} \tag{19}$$

Computing the partial derivatives:

$$\frac{\partial G}{\partial c} = -q'(R)$$

$$\frac{\partial G}{\partial R} = q''(R)(R - c) + q'(R) + q'(R) = q''(R)(R - c) + 2q'(R)$$

Therefore:

$$\frac{dR^*}{dc} = -\frac{-q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} = \frac{q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} \tag{20}$$

From the first-order condition (16), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$\frac{dR^*}{dc} = \frac{q'}{q''\left(-\frac{q}{q'}\right) + 2q'} = \frac{q'}{-\frac{q\,q''}{q'} + 2q'} = \frac{(q')^2}{-(q)(q'') + 2(q')^2} \tag{21}$$

Dividing numerator and denominator by $(q')^2$:

$$\frac{dR^*}{dc} = \frac{1}{2 - \frac{q(R^*)\,q''(R^*)}{[q'(R^*)]^2}} \tag{22}$$

Since $c = (1 - p_e)T$ and hence $dc/dT = (1 - p_e)$, the pass-through rate with respect to the tax liability is:

$$\boxed{\rho = \frac{dR^*}{dT} = \frac{dR^*}{dc} \cdot \frac{dc}{dT} = \frac{1 - p_e}{2 - \theta}}, \tag{23}$$

where $\theta \equiv \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by Weyl and Fabinger (2013).

In the pass-through formula (23), we require $\theta < 2$ for $\rho \geq 0$. This restriction follows from the second-order condition of the landlord's maximization problem, which ensures that the optimal rent R^* is a well-defined interior maximum. To see this, note that the second-order condition requires:

$$\frac{d^2}{dR^2}\left[q(R)(R - c)\right] = q''(R)(R - c) + 2q'(R) < 0 \tag{24}$$

From the first-order condition (16), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$q''(R^*)\left(-\frac{q(R^*)}{q'(R^*)}\right) + 2q'(R^*) = -\frac{q(R^*)\, q''(R^*)}{q'(R^*)} + 2q'(R^*) \tag{25}$$

Factoring out $q'(R^*)$:

$$= q'(R^*)\left(2 - \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}\right) = q'(R^*)\,(2 - \theta) \tag{26}$$

Since $q'(R^*) < 0$, the expression in (26) is negative if and only if $2 - \theta > 0$, i.e., $\theta < 2$. When $\theta \geq 2$, the landlord's profit function is convex at the candidate solution, and no interior optimum exists.